Week 9 Notes

Critical Points:
Maxima and Minima, Areas, Riemann Sums and Integration

Course: Mathematics–1

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Lecture 1 — Critical Points: Local Maxima and Minima

Intervals of increase and decrease; turning points; horizontal tangents; critical and saddle points; the second derivative test

1 Recalling intervals of increase and decrease

Before discussing critical points, let us recall some things we have already studied. For a function $f(x)$ we have met the notion of an **interval of increase** and an **interval of decrease**. An interval of increase is an interval on which the function $f(x)$ is increasing, and an interval of decrease is one on which it is decreasing.

Intervals of increase and decrease

Let I be an interval contained in the domain of f.

$$\textbf{Increase:} \quad x_1, x_2 \in I,\ \ x_1 < x_2 \implies f(x_1) < f(x_2).$$

$$\textbf{Decrease:} \quad x_1, x_2 \in I,\ \ x_1 < x_2 \implies f(x_1) > f(x_2).$$

So on an interval of increase, x_1 being less than x_2 means $f(x_1)$ is less than $f(x_2)$; an interval of decrease is exactly the opposite — the function decreases as we move from x_1 to x_2.

2 Turning points, local maxima and local minima

We have also met the notion of a **turning point**.

Definition 1.1 — Turning point

A point x is a **turning point** of f if one of the following two phenomena takes place:

1. there is an interval of increase **ending** at x and an interval of decrease **beginning** at x; or
2. there is an interval of decrease **ending** at x and an interval of increase **beginning** at x.

Note carefully what phenomenon 1 says: the function need not increase throughout up to x. There is *some* interval — it could be small, it could be large — which ends at x, and then from x another interval begins on which the function decreases.

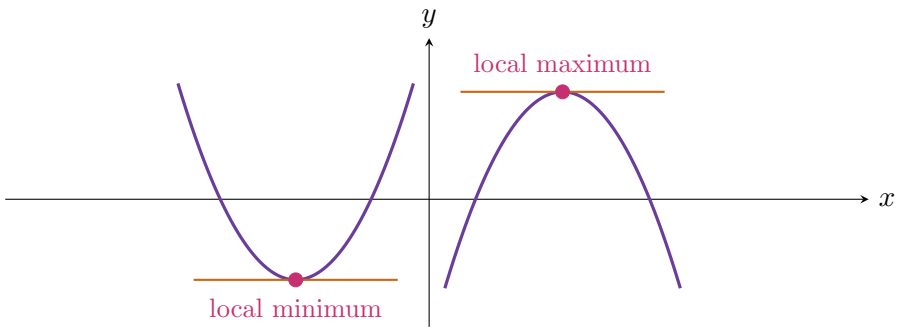
Let us give these two phenomena new names.

Definition 1.2 — Local maximum and local minimum

- A point $a \in \mathrm{Dom}(f)$ is a **local maximum** of a function f if there exists a $\delta > 0$ such that $f(a) \geq f(x)$ for all $x \in \mathrm{Dom}(f)$ satisfying $|x - a| < \delta$.
- A point $a \in \mathrm{Dom}(f)$ is a **local minimum** of a function f if there exists a $\delta > 0$ such that $f(a) \leq f(x)$ for all $x \in \mathrm{Dom}(f)$ satisfying $|x - a| < \delta$.

A note on plurals

The plural of *maximum* is **maxima**, and the plural of *minimum* is **minima**.



In each case the orange segment is the **tangent** at the turning point — and in each case it is **horizontal**.

3 Example 1: a degree-six polynomial

Example 1.1

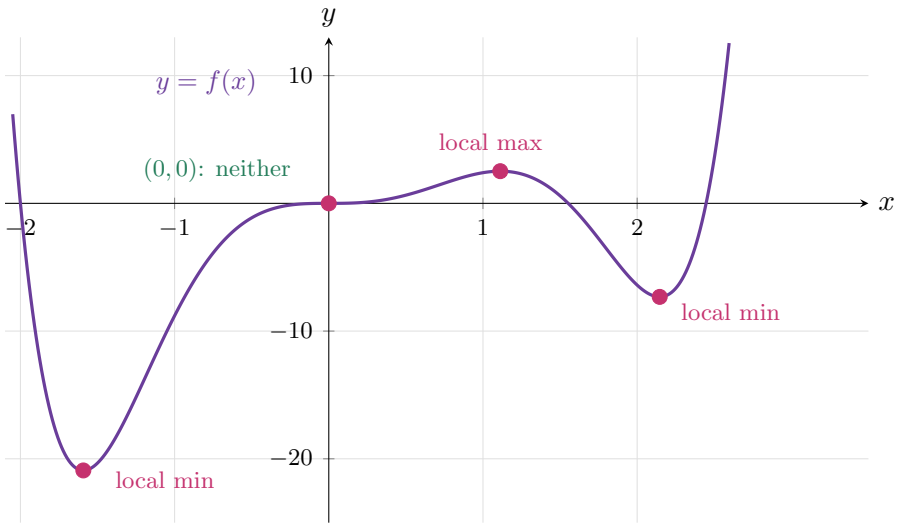
Consider

$$f(x) = \left(x^2 - 4x + 3.8\right)(x + 2)\,x^3.$$

What are its intervals of increase and decrease, and where are its local maxima and minima?

Multiplied out, this is the polynomial

$$f(x) = x^6 - 2x^5 - \frac{21}{5}x^4 + \frac{38}{5}x^3.$$



Reading the graph from the left:

- The whole portion coming down from above is an **interval of decrease**.
- After that, up to approximately $x = 1.11$, we have an **interval of increase** — although in the middle, close to 0, something strange is happening.
- Then there is again an **interval of decrease**, down to about $x = 2.15$.
- After that we have **increase**, and it probably increases all the way beyond.

Now, what are these points?

Point	Value	What it is
$x \approx -1.59$	$f \approx -20.91$	local minimum — and in fact the **global** minimum
$x = 0$	$f = 0$	neither a maximum nor a minimum (see below)
$x \approx 1.11$	$f \approx 2.52$	local maximum
$x \approx 2.15$	$f \approx -7.32$	local minimum

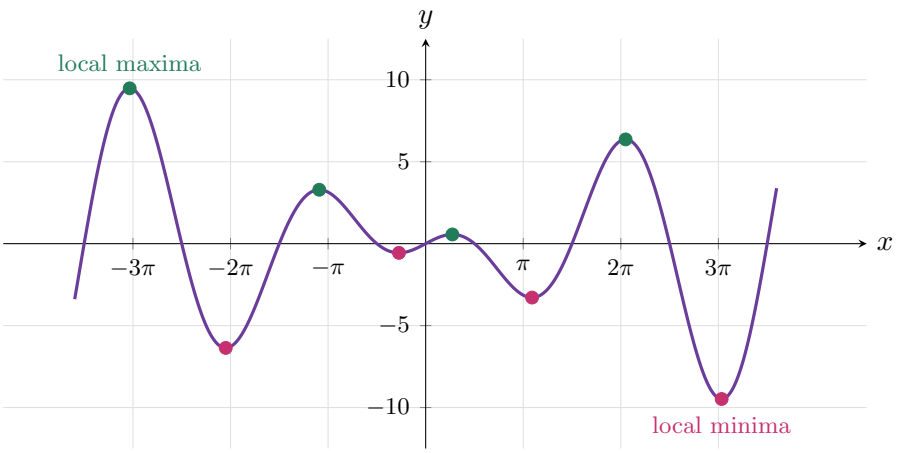
Why is the point at $x \approx 2.15$ a local minimum? Because there is a **valley**, or a trough, there: if we choose some small interval around it, the value taken at that point is the smallest on that interval. It is a local minimum, but it is clearly *not* a global minimum. The point at $x \approx -1.59$ is also a local minimum on a small interval around it, and there it so happens that the value *is* the global minimum — the smallest among all possible values of $f(x)$.

And at $x \approx 1.11$ we have a local maximum. Here we must be careful about the interval: a small interval around the point will do, or a somewhat larger one, but we cannot take the interval $(-2.5, 2.5)$, because on that interval there are values which are larger.

4 Example 2: $f(x) = x \cos x$

Example 1.2

Locate the local maxima and local minima of $f(x) = x \cos x$.



Marking the peaks gives one local maximum, then another, then another, and so on; marking the troughs gives one local minimum after another. Since this is a nice wave-like function with **increasing amplitude**, there are lots of local maxima — in fact **infinitely many** — and, by the same reasoning, infinitely many local minima as well.

Definition 1.3 — Local extremum

A **local extremum** is either a local minimum or a local maximum — it is an extreme value for a small interval around that point.

Such values are turning points, and they often carry a great deal of information about the function. So the question we want to answer is: **how do we find the local extrema?**

5 At a turning point the tangent is horizontal

One way to think about finding them is to look at the **tangents** — we have been studying tangents for a while, and this is another application of them.

The observation

In both examples, the tangents at the turning points are **horizontal** — flat, that is, parallel to the x-axis.

In the first example, drawing the tangent at the left-hand local minimum, at the local maximum, and at the right-hand local minimum, each one comes out horizontal — all three points that we identified as local extrema behave the same way. In the second example, drawing the tangent at any of the peaks or troughs of $x\cos x$ likewise gives a horizontal line. The picture is very clear and unambiguous.

But how do we make this **formal**? The way to do it is to introduce **derivatives**, as we have been doing: derivatives give us an algebraic way of getting hold of equations, and that will let us first prove that the tangent is horizontal, and then use the fact for the further study of local maxima and minima.

5.1 What we already know about tangents

Recall from the previous lecture:

$$\text{the tangent to } f \text{ at } a \text{ exists and is not vertical} \iff f \text{ is differentiable at } a,$$

and in that case the equation of the tangent is

$$y = f'(a)\,(x-a) + f(a).$$

Hence proving that the tangent (if it exists) is horizontal at a turning point is **equivalent** to showing that

$$f'(a) = 0.$$

6 The theorem and its proof

Theorem 1.1

If a is a turning point of f and $f'(a)$ exists, then

$$f'(a) = 0.$$

Note that we cannot guarantee that the derivative exists — tangents may not exist in general — so we assume it does.

Proof

Suppose, for the sake of argument, that a is a **local maximum**. What does that mean? It means that there is some open interval

$$(a - \epsilon,\ a + \epsilon)$$

such that, whenever $0 < |h| < \epsilon$,

$$f(a - h) < f(a) \qquad \text{and} \qquad f(a + h) \leq f(a).$$

That is exactly when we say a is a local maximum.

Now apply the definition of the derivative:

$$f'(a) = \lim_{h \to 0} \frac{f(a + h) - f(a)}{h}.$$

The limit from the right

Take $0 < h < \epsilon$. Then $f(a + h) < f(a)$, so the **numerator is negative** while the denominator h is positive. Hence the whole quotient is negative, and

$$f'(a) = \lim_{h \to 0^+} \frac{f(a + h) - f(a)}{h} \leq 0.$$

The limit from the left

On the other hand, $f'(a)$ also equals the limit from the left. Here $-\epsilon < h < 0$, so the numerator $f(a + h) - f(a)$ is again negative — but the denominator h is **also negative**, and negative divided by negative gives a positive number. Hence

$$f'(a) = \lim_{h \to 0^-} \frac{f(a + h) - f(a)}{h} \geq 0.$$

So we know that $f'(a)$ exists, that $f'(a) \geq 0$, and that $f'(a) \leq 0$. Together these force

$$\boxed{f'(a) = 0}$$

and one can see here the power of the left limit and the right limit working together. Since $f'(a) = 0$ says exactly that the tangent is horizontal — parallel to the x-axis — the theorem is proved. ■

7 Critical points

Definition 1.4 — Critical point

A point a is called a **critical point** of a function f if

- f is **not differentiable** at a; **or**
- $f'(a) = 0$.

Every turning point is a critical point

If f is differentiable at a turning point a, then $f'(a) = 0$ by Theorem 1.1, so a is a critical point. And if f is not differentiable at a, then a is a critical point anyway.

So in order to find turning points — that is, local maxima and local minima — we shall **first find the critical points**.

8 Is every critical point a turning point?

Suppose f is differentiable. Among the critical points, some will be local maxima and some will be local minima — but are there any others? If there were none, then by finding the set of critical points we would have the entire set of local maxima and minima.

Unfortunately, **that is not the case**.

Look again at

$$f(x) = \left(x^2 - 4x + 3.8\right)(x + 2)\,x^3$$

at the value $x = 0$. Since the expression carries a factor x^3, differentiating gives $f'(0) = 0$, so $x = 0$ **is** a critical point — yet the graph shows it is neither a local maximum nor a local minimum. (Working out $f'(0) = 0$ from the expression is a good exercise.)

Definition 1.5 — Saddle point

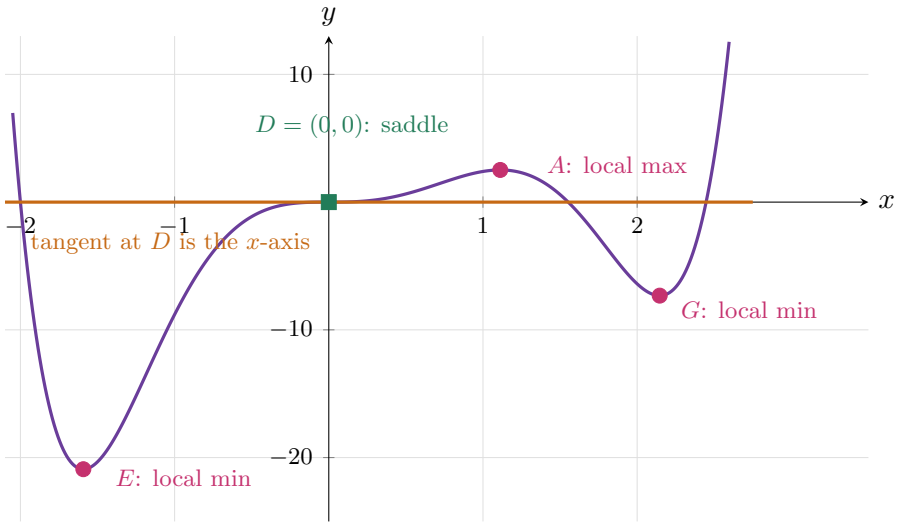
A **saddle point** is a critical point which is neither a local maximum nor a local minimum.

A caveat about the name

Different sources use different names for this. The word *saddle point* is common once we go beyond one dimension, which we will do in subsequent weeks. In one dimension other words are used — some sources say **inflection point** — so be careful, depending on which source you study.

8.1 The picture with all critical points marked

Here is the same function again, now with all four critical points marked, together with the tangent at the saddle point.



If we draw the tangent to f at the point $(0,0)$, it is precisely the x-axis — and this can be checked by writing down the equation, since differentiating the x^3 factor gives $f'(0) = 0$.

8.2 What is special about the saddle point

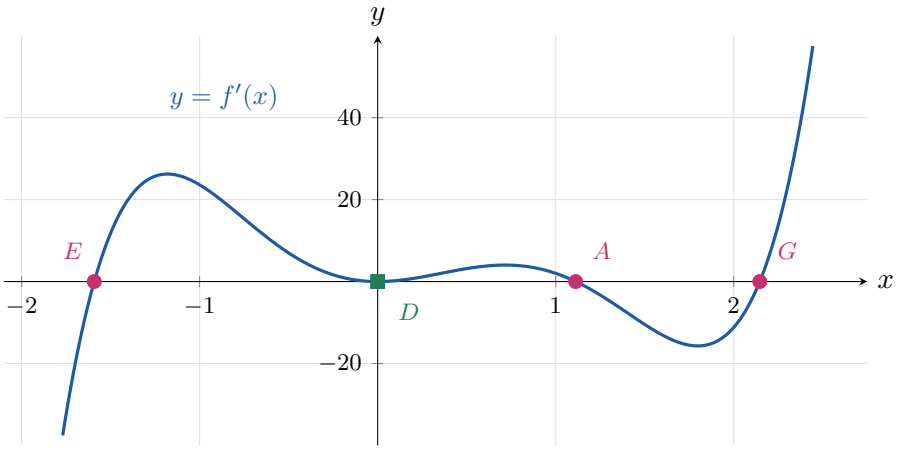
At a saddle point the function behaves in a slightly odd way. It is not a local minimum or maximum, but there is a change of shape:

> Just to the **left** of D the curve is **concave below**, and just to the **right** of D it is **concave above**. So there is a change of concavity occurring *at* D, and that change is exactly what this point reflects. Whenever such a thing happens, the point is a **saddle point**.

8.3 Looking at the derivative

Let us also draw f'. Differentiating,

$$f'(x) = 6x^5 - 10x^4 - \frac{84}{5}x^3 + \frac{114}{5}x^2 = x^2\left(6x^3 - 10x^2 - 16.8x + 22.8\right).$$



These four points are exactly the **zeros of** f', so we have identified all the critical points of this function correctly. But notice how differently they behave:

- At E, the value of f' is 0 and f' is nicely **increasing** there.
- At A, the value of f' is 0 and f' is **decreasing** there.
- At G, again f' is 0 and f' is **increasing**.
- At D, f' behaves in a totally different way: D is actually a **local minimum of the derivative** f'. Whenever that happens, the point is a saddle point.

9 The second derivative test

How do we decide which critical points are local maxima and which are local minima? Assume f is differentiable — indeed **twice** differentiable — and use the **second derivative test**.

The idea behind the test

Just as the first derivative f' checks the **monotonicity of** f, the second derivative f'' checks the **monotonicity of** f'. From that monotonicity we can read off the behaviour of f at the critical point — exactly as we just did by eye on the graph of f'.

The second derivative test

Let f be twice differentiable and let a be a critical point. Then:

1. If $f''(a) > 0$, then a is a **local minimum**.
2. If $f''(a) < 0$, then a is a **local maximum**.
3. If $f''(a) = 0$, then the test is **inconclusive**.

The third case is a little disappointing — the test is not an absolute one — and we shall see later why that is so.

9.1 Applying the test to Example 1.1

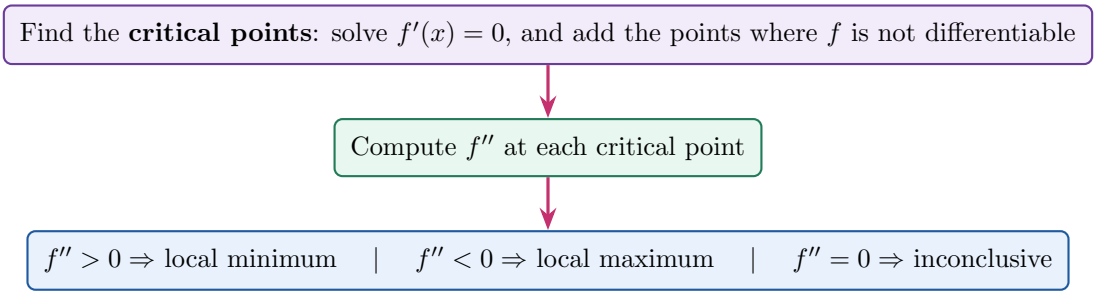
For $f(x) = x^6 - 2x^5 - \frac{21}{5}x^4 + \frac{38}{5}x^3$ we have

$$f''(x) = 30x^4 - 40x^3 - \frac{252}{5}x^2 + \frac{228}{5}x.$$

Critical point	f'	f''	Conclusion
E: $x \approx -1.59$	0	$\approx +153.7 > 0$	local minimum
D: $x = 0$	0	$= 0$	test inconclusive — here a **saddle point**
A: $x \approx 1.11$	0	$\approx -20.8 < 0$	local maximum
G: $x \approx 2.15$	0	$\approx +106.9 > 0$	local minimum

Notice that the saddle point is exactly the case where f'' vanishes and the test tells us nothing — which is why the graph of f', and the change of concavity, had to be inspected directly.

10 Summary of the procedure



So: we obtain the critical points by setting the derivative to 0 (or by finding where the function fails to be differentiable); and then, assuming the function is twice differentiable, the second derivative test tells us that a negative second derivative gives a local maximum, a positive one gives a local minimum, and a zero leaves the matter inconclusive.

Lecture 2 — Computing Maxima and Minima

Worked examples of the second derivative test; why the test works; closed intervals and global extrema

1 Recalling the previous lecture

We continue from the previous video, where we discussed critical points and computing maxima and minima. Let us quickly recall what we studied.

What we already have

- A point a is a **critical point** of f if either f is not differentiable at a, or $f'(a) = 0$.
- A **saddle point** is a critical point which is not a local maximum or a local minimum — that is, not a turning point. Remember that turning points are the local maxima and local minima; a saddle point is one where the derivative is 0 but it is not a turning point (or where the function may not be differentiable).
- **Second derivative test.** If f is twice differentiable and a is a critical point:

  $f''(a) > 0 \Rightarrow$ local minimum, $\quad f''(a) < 0 \Rightarrow$ local maximum, $\quad f''(a) = 0 \Rightarrow$ inconclusive.

Let us now apply the test to some examples and see how it is used.

2 Example 1: $f(x) = x^3 - 12x$

Example 2.1

Find and classify the critical points of $f(x) = x^3 - 12x$.

This is a nice function, so compute the derivative and set it to 0:

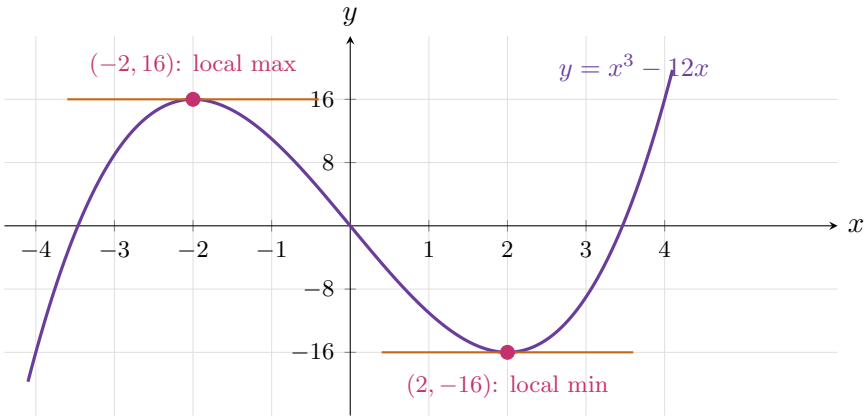
$$f'(x) = 3x^2 - 12 = 0 \implies x^2 = 4 \implies x = \pm 2.$$

So the **critical points** are $x = 2$ and $x = -2$.

Now check the second derivative, $f''(x) = 6x$:

$$f''(2) = 12 > 0, \qquad f''(-2) = -12 < 0.$$

Therefore $x = 2$ is a **local minimum**
and $x = -2$ is a **local maximum**.



3 Example 2: $f(x) = \cos x$

Example 2.2

Find and classify the critical points of $f(x) = \cos x$.

We repeat the same procedure. Here

$$f'(x) = -\sin x,$$

and we know the zeros of $\sin x$ — they are the multiples of π. So setting $f'(x) = 0$, the critical points are

$$x = k\pi, \qquad k \in \mathbb{Z}.$$

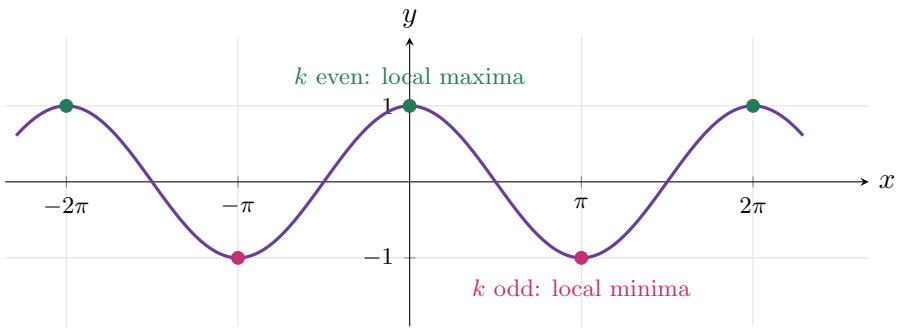
Now compute $f''(x) = -\cos x$ and evaluate at $k\pi$:

$$f''(k\pi) = -\cos(k\pi),$$

and this depends on the **parity of** k. Since $\cos 0 = 1$, $\cos 2\pi = 1$, and so on,

$$f''(k\pi) = \begin{cases} -1 < 0, & k \text{ even} \quad (k \equiv 0 \bmod 2), \\ +1 > 0, & k \text{ odd}. \end{cases}$$

$k\pi$ with k **even** are **local maxima**; $k\pi$ with k **odd** are **local minima**.



4 Example 3: $f(x) = x^3 + x^2 - x + 5$

Example 2.3

Find and classify the critical points of $f(x) = x^3 + x^2 - x + 5$.

Differentiating,

$$f'(x) = 3x^2 + 2x - 1.$$

This looks slightly complicated, but it is a **quadratic**, so setting it to 0 we know the roots. Using the quadratic formula with $a = 3$, $b = 2$, $c = -1$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-2 \pm \sqrt{4 - 4(3)(-1)}}{2(3)} = \frac{-2 \pm \sqrt{4 + 12}}{6} = \frac{-2 \pm \sqrt{16}}{6} = \frac{-2 \pm 4}{6}.$$

So the roots are

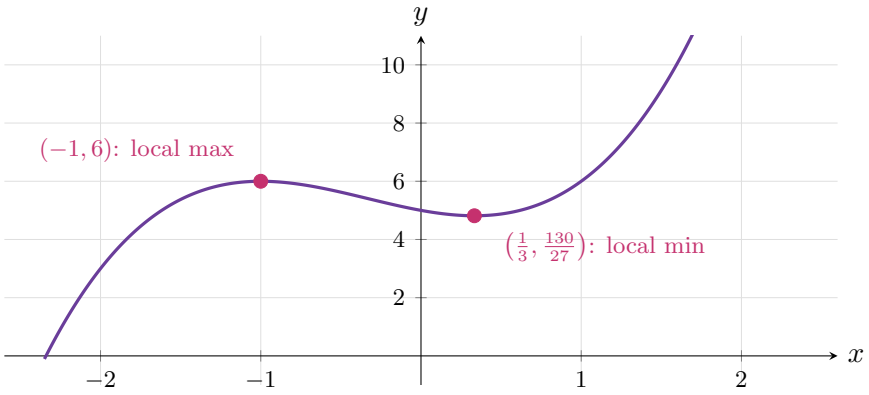
$$\frac{-2-4}{6} = -1 \qquad \text{and} \qquad \frac{-2+4}{6} = \frac{1}{3},$$

and these are the critical points.

Now $f''(x) = 6x + 2$, so

$$f''(-1) = -6 + 2 = -4 < 0 \implies \textbf{local maximum},$$

$$f''\left(\tfrac{1}{3}\right) = 6 \cdot \tfrac{1}{3} + 2 = 2 + 2 = 4 > 0 \implies \textbf{local minimum}.$$



So the method is clear: this is how we check what the local minima and maxima are. It should be pointed out, though, that sometimes the second derivative test **does not work** — if it is inconclusive, one has to work harder to decide which points are local maxima and minima.

5 Why does the test work?

Let us see what happens to the **tangents** as we move close to a critical point.

At a local minimum

Watch the tangent as it slides through a small neighbourhood of the point. Initially the slope — the derivative value — is **negative**; at the point itself it becomes 0, i.e. the tangent is horizontal; and then it becomes **positive**.

$$f' : \text{ negative } \longrightarrow 0 \longrightarrow \text{ positive}$$

So f' is **increasing** through the point, and therefore the second derivative — which measures the monotonicity of f' — is **positive**.

At a local maximum

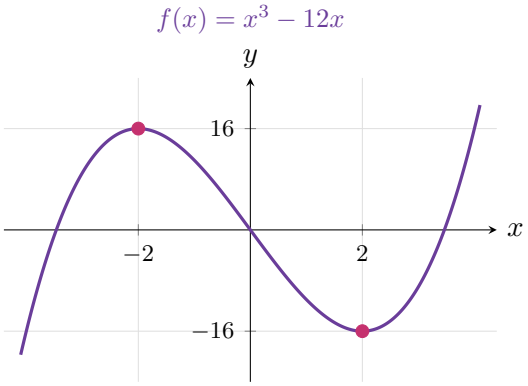
The same phenomenon occurs in reverse: near the point f' is **decreasing**, so f'' is **negative**.

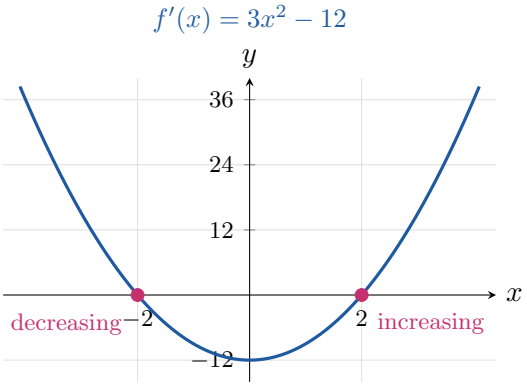
At a saddle point

Here something strange happens. Following the derivative through the point, it is decreasing, decreasing, decreasing up to 0 — and as soon as it hits 0 it starts **increasing** again. So around that neighbourhood f' is *neither* strictly increasing *nor* strictly decreasing, and that is exactly why f'' is 0 there.

Choose a small enough neighbourhood

The word **local** matters. If the interval chosen around the point is too large, f' need not be monotonic on it: somewhere along the graph of f' there is an intermediate point at which f' turns around — stopping increasing and starting to decrease, or the other way about — and that is precisely where f'' becomes 0. So a small interval around the point must be chosen; on such an interval f' will be strictly monotonic and the reasoning goes through.





On the right, f' is *decreasing* at $x = -2$ — hence $f''(-2) < 0$ and that point is a local maximum — and *increasing* at $x = 2$ — hence $f''(2) > 0$ and that point is a local minimum.

6 Local extrema on a closed interval

Sometimes we want the local extrema of a function on a **closed interval** $[a, b]$. In that case, in addition to finding the extrema by the method already described — setting $f'(x) = 0$ and so on — it is possible that the **end points** are also local extrema, and they must be considered separately.

Example 2.4

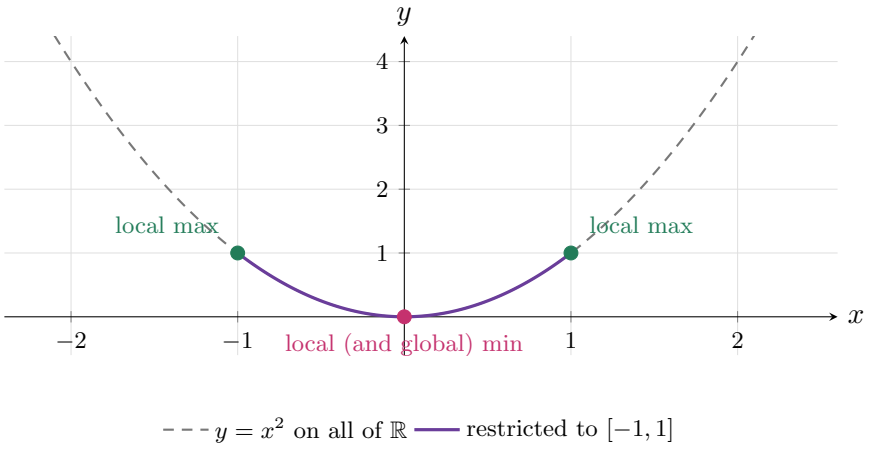
Consider $f(x) = x^2$ on the interval $[-1, 1]$.

Computing the first derivative and setting it to 0:

$$f'(x) = 2x = 0 \implies x = 0.$$

And $f''(x) = 2$, so $f''(0) = 2 > 0$ and $x = 0$ is a **local minimum** — in fact the **global minimum**, as we know from the graph of x^2.

But if we consider the function *only* on this interval, then $x = -1$ and $x = 1$ are also **local maxima**. Why? Because a neighbourhood now means only points lying within $[-1, 1]$, and on such a small neighbourhood of the end point, the value at the end point is the largest.



For **closed intervals** you must also consider the **end points** separately.

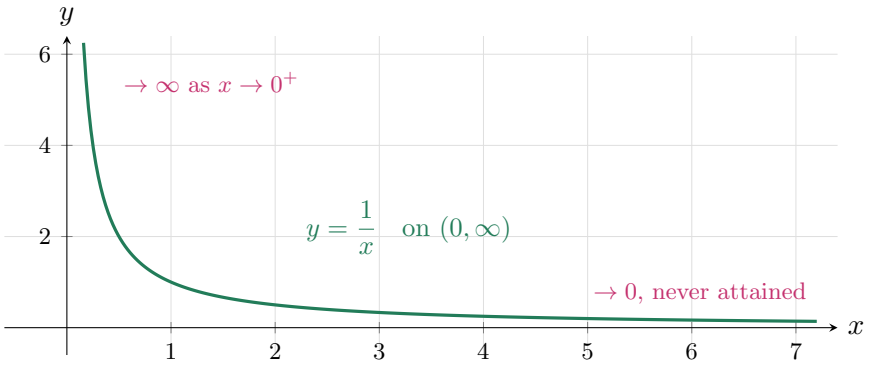
7 Global maxima and minima

Sometimes we want the maximum or minimum value of a function on a particular interval I — and here by maximum or minimum we mean the **global** maximum or minimum. Now, this **may not exist** in general: sometimes it does and sometimes it does not.

Example 2.5

Consider $f(x) = \dfrac{1}{x}$ on the interval $(0, \infty)$.

This function has **neither a global maximum nor a global minimum**. Why? Because it approaches infinity as we come close to 0, and it approaches 0 as we go towards infinity — but it never attains the value 0, nor of course does it attain the value infinity, which is not a real number at all. So it can never attain a maximum or a minimum value. In this case there is in fact no local maximum or local minimum either.



So in general, global maxima and global minima **may not exist**. But here is a fact.

When the maximum and minimum must exist

If the interval I is **closed** (it includes its boundary points) and **bounded** (it does not go off to infinity — it lies within some $[a, b]$), **and** the function f is **continuous**, then the maximum and the minimum **must exist**.

The continuity assumption is important. Note also that the maximum and minimum are in particular local maxima or local minima — *unless* they occur at boundary points, in which case they need not be local maxima or minima in the sense we studied earlier.

> To find the maximum and minimum: find the **critical points** and the **boundary points**, and check the value of f at all of them.

In fact this can be done for any function which is defined **piecewise continuously**, with finitely many pieces, on a closed and bounded interval.

8 A piecewise example

Example 2.6

Let

$$f(x) = \begin{cases} x^3 + x^2 - x + 5, & 0 \le x \le 100, \\ x^3 + 2x^2 + x - 5, & -100 \le x < 0. \end{cases}$$

Find the global maximum and the global minimum.

Step 1: the boundary points

The boundary points are 100 and -100; and we must **also include** 0 as a boundary point, because that is where the definition of the function changes.

$$\text{boundary points:} \qquad -100, \quad 0, \quad 100.$$

Step 2: the critical points on each piece

When computing critical points on a piece, only those lying **inside the open interval** for that piece count — not even its boundary points. Differentiating,

$$f'(x) = \begin{cases} 3x^2 + 2x - 1, & 0 < x < 100, \\ 3x^2 + 4x + 1, & -100 < x < 0. \end{cases}$$

First piece. Solving $3x^2 + 2x - 1 = 0$ by the quadratic formula,

$$x = \frac{-2 \pm \sqrt{4 - 4(3)(-1)}}{2(3)} = \frac{-2 \pm \sqrt{16}}{6} = \frac{-2 \pm 4}{6} \implies x = -1 \text{ or } x = \frac{1}{3}.$$

But -1 is **not** in the interval $(0, 100)$, so we must not take it. The only critical point from the first piece is $x = \frac{1}{3}$.

Second piece. Solving $3x^2 + 4x + 1 = 0$,

$$x = \frac{-4 \pm \sqrt{16 - 4(3)(1)}}{2(3)} = \frac{-4 \pm \sqrt{4}}{6} = \frac{-4 \pm 2}{6} \implies x = -1 \text{ or } x = -\frac{1}{3}.$$

Both of these lie inside $(-100, 0)$, so both count.

> **Critical points:** $\frac{1}{3}, \quad -\frac{1}{3}, \quad -1$ **Boundary points:** $-100, \quad 0, \quad 100$

Step 3: evaluate f at all six points

$$f\left(\tfrac{1}{3}\right) = \frac{1}{27} + \frac{1}{9} - \frac{1}{3} + 5 = \frac{1 + 3 - 9 + 135}{27} = \frac{130}{27} \approx 4.81,$$

which is close to 5 — slightly less, between 4 and 5. Similarly, using the *second* expression (note the 2 multiplying x^2),

$$f\left(-\tfrac{1}{3}\right) = -\frac{1}{27} + \frac{2}{9} - \frac{1}{3} - 5 = \frac{-1 + 6 - 9 - 135}{27} = -\frac{139}{27} \approx -5.15,$$

which lies between -5 and -6. Next,

$$f(-1) = -1 + 2 - 1 - 5 = -5, \qquad f(0) = 5.$$

Finally the two far end points:

$$f(100) = 100^3 + 100^2 - 100 + 5 = 1\,009\,905,$$

$$f(-100) = (-100)^3 + 2(100)^2 - 100 - 5 = -980\,105.$$

We need not labour over these values: it is obvious that they are **really large** compared with the others, which are all in the range of about ± 5. The first is of the order 100^3 and is clearly larger than all the rest; the second is of order -100^3, so in absolute value it is very large and carries a negative sign, making it the smallest of all.

Point	Type	f	Conclusion
$x = -100$	boundary	$-980\,105$	**global minimum**
$x = -1$	critical	-5	
$x = -\frac{1}{3}$	critical	$-\frac{139}{27} \approx -5.15$	
$x = 0$	boundary	5	
$x = \frac{1}{3}$	critical	$\frac{130}{27} \approx 4.81$	
$x = 100$	boundary	$1\,009\,905$	**global maximum**

So among all the values taken by the function, the one at -100 is the smallest and the one at 100 is the largest.

Step 4: classifying the critical points

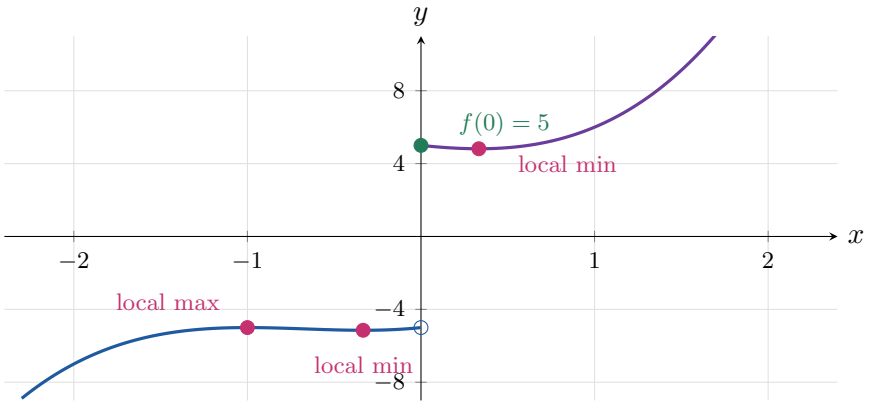
We can also go ahead and ask what kind of points the critical points are. For that we compute f'' — again only on the open intervals:

$$f''(x) = \begin{cases} 6x + 2, & 0 < x < 100, \\ 6x + 4, & -100 < x < 0. \end{cases}$$

Plugging in the values and checking the signs (hopefully they are non-zero, so the test applies):

$$f''\left(\tfrac{1}{3}\right) = 4 > 0, \qquad f''\left(-\tfrac{1}{3}\right) = 2 > 0, \qquad f''(-1) = -2 < 0.$$

So $\frac{1}{3}$ and $-\frac{1}{3}$ are **local minima**, and -1 is a **local maximum**.



The picture shows the region near the break point; the two far boundary values, being of order $\pm 10^6$, are far off the scale and are where the global extrema actually occur.

9 A warning example

Example 2.7

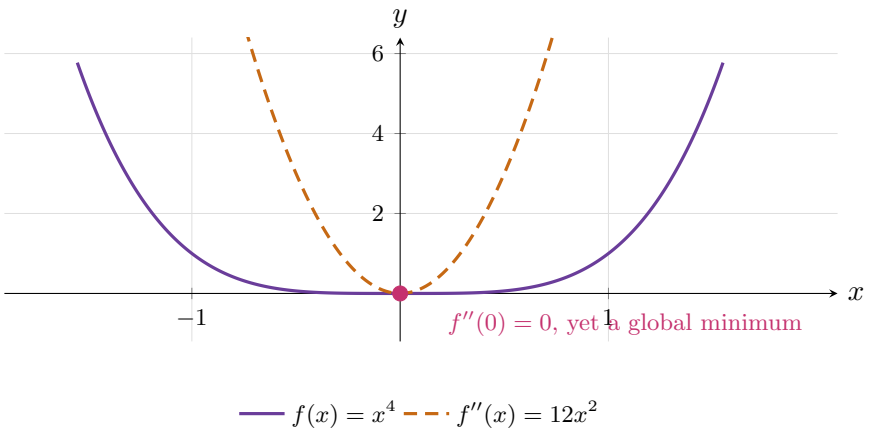
Consider $f(x) = x^4$.

Clearly 0 is the **global minimum** — and in particular a local minimum — so we would hope that the test shows this up. Let us see what it gives:

$$f'(x) = 4x^3 = 0 \implies x = 0 \quad \text{(the critical point)},$$

$$f''(x) = 12x^2 \implies f''(0) = 0.$$

So the second derivative test ends up being **inconclusive** — even though the function is one we can analyse clearly and obviously.



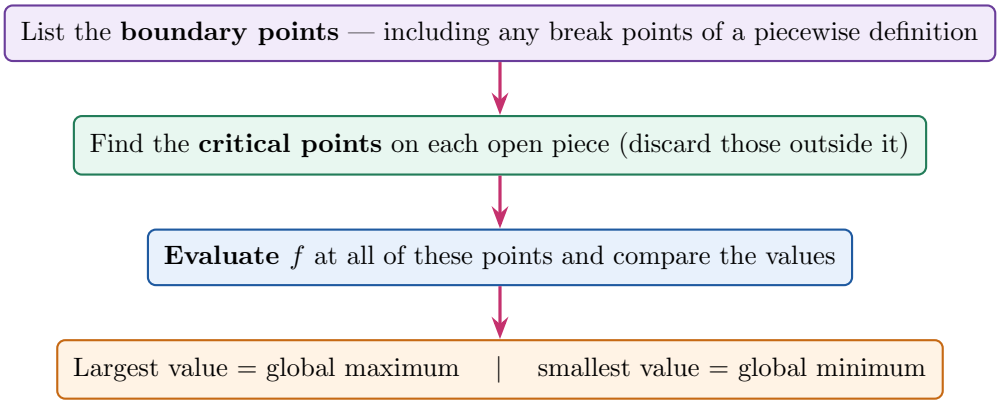
So be careful: when we land in the inconclusive situation, the point may still be a local — indeed a **global** — minimum or maximum. That has to be analysed by other methods.

What the inconclusive case can hide

The inconclusive situation ($f''(a) = 0$) can arise from a local minimum, a local maximum, *or* a saddle point — any of them. To decide which, we have to check what happens to the **derivative**

function close to that point.

10 The procedure on a closed and bounded interval



So: on a closed and bounded interval, or for a piecewise continuous function, we must also check the function values **at the boundary points** and at the break points where the definition changes. Only then can we conclude about the local or global minimum and maximum.

Lecture 3 — Computing Areas

From the rectangle to polygons; Archimedes and the circle; divide, conquer and take limits

1 Where we are going

In this lecture we talk about **computing areas**. Much of it is a recollection of things learnt in school, but towards the end we look at shapes that are a little more involved, with the idea of evolving a *method* for computing their areas. That will set the tempo for the next lectures, where a more detailed version of the same method leads us to what is called the **theory of integration**.

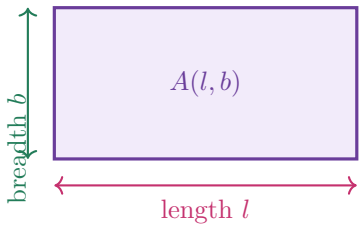
2 The area of a rectangle

A rectangle is described by its **length** and its **breadth**. If we know both, the rectangle is unique — any two rectangles with the same length and breadth can be superimposed on one another, and in particular they have the same area.

So the area is a **function** of the length and the breadth, and we call it

$$A(l, b).$$

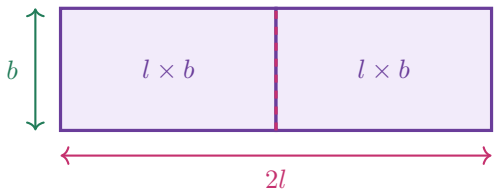
Everybody knows what this function is; the point here is to understand *why* we get that formula.



2.1 Doubling the length

What happens if we **double the length**? Divide the new rectangle down the middle: we get two rectangles, each with length l and breadth b, so each has the same area as the original. Hence the total area is twice the original:

$$A(2l,\, b) = 2\, A(l,\, b).$$



The same argument holds for 3 times, 4 times, 20 times, 100 times, and so on. In other words, for a natural number n,

$$A(n\, l,\, b) = n\, A(l,\, b), \qquad n \in \mathbb{N}$$

— we take n natural since we do not want to talk about negative areas.

2.2 Multiplying by a rational number

What if we multiply by a **rational** number — say we take one and a half times the length? The formula still holds: one and a half is $\frac{3}{2}$, so we divide the original into halves and take three of them. Each half has area $\frac{1}{2}A(l,b)$, so the total is $\frac{3}{2}A(l,b)$.

The same idea works for any rational number:

$$A\Big(\frac{m}{n}\,l,\ b\Big) = \frac{m}{n}\,A(l,b), \qquad m,n \in \mathbb{N}.$$

And there is nothing special about the length — the same holds for the breadth. So for rational c and d,

$$A(c\,l,\ d\,b) = c\,d\,A(l,b).$$

2.3 From rationals to reals

Once a formula holds for all rational numbers, we can do what we have done before: **take limits**. Choosing a sequence of rationals tending to any given real number and passing to the limit on both sides, we conclude that the formula holds for *any* two positive real numbers c and d:

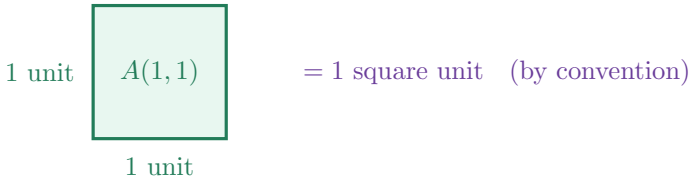
$$\boxed{A(c\,l,\ d\,b) = c\,d\,A(l,b), \qquad c,d>0}$$

2.4 The punchline

Now take $c = l$ and $d = b$ themselves. Writing $l = l \times 1$ and $b = b \times 1$,

$$A(l,b) = A(l\cdot 1,\ b\cdot 1) = l\,b\,A(1,1).$$

And what is $A(1,1)$? It is the area of the rectangle with 1 unit as length and 1 unit as breadth — that is, a **unit square**.



So the area of a rectangle reduces to

$$\text{area} = \text{length} \times \text{breadth} \times \big(\text{area of the unit square}\big),$$

and by **convention** we take that last factor to be 1, giving $l \times b$ square units.

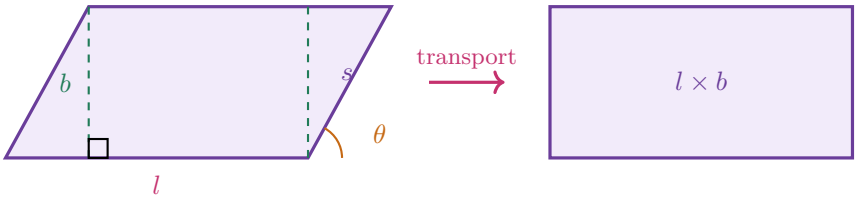
Why units matter

If we change the units, $A(1,1)$ changes correspondingly — and that is exactly why areas must always be written *in terms of some unit*. Suppose we suddenly decide to measure in centimetres: we must ask how many centimetres the original unit is, convert, and we may not get 1 square centimetre but some constant, which is precisely $A(1,1)$ in the new units. That constant then multiplies $l \times b$, where l and b are still in the original units.

This kind of explanation is not usually given in school, where the formula is simply handed to us and seems perfectly correct — but what we have described here is a kind of **proof**, and it also tells us why the units have to be there.

3 The parallelogram

Once we can measure rectangles, we can ask about other figures. For the parallelogram there is a **trick**: drop a perpendicular at each end, and **transport** the right-hand triangle over to the left. What remains is a **rectangle** of exactly the same area.



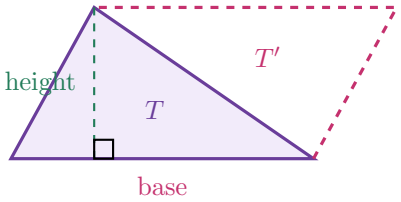
The length of the rectangle is the same as the length l of the parallelogram's base. The breadth is the length of the perpendicular we dropped. Often we are not given that perpendicular directly, but rather the angle θ and the length s of the slanted side. Then, from trigonometry, the opposite side has length $s \sin\theta$, so

$$b = s\,\sin\theta \qquad \Longrightarrow \qquad \boxed{\text{Area of parallelogram} = l\ s\ \sin\theta}$$

since the area is $l \times b$.

4 The triangle

For the triangle there is another trick: **complete it to a parallelogram** by drawing the missing half. We then have two triangles of the *same* area, so the parallelogram's area is twice the triangle's.



$\text{area}(T) = \text{area}(T')$

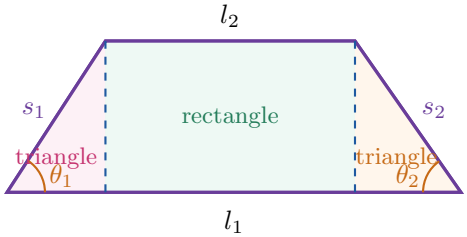
Hence the area of the triangle is half the area of the parallelogram, and since we know how to compute that,

$$\boxed{\text{Area of triangle} = \frac{1}{2} \times \text{base} \times \text{height}}$$

And of course, if instead of the height we are given a side s and an angle θ, we again use $s \sin\theta$, giving $\frac{1}{2}\, l\, s \sin\theta$.

5 The trapezium

A trapezium is a little more complicated. Drop the two perpendiculars, and the figure breaks up into **three pieces** — a rectangle in the middle and a triangle at each end — all of which we know how to handle.



The only remaining question is how to describe a trapezium, i.e. which lengths and angles we are given. The possible data are

$$l_1, \quad l_2, \quad \theta_1, \quad \theta_2, \quad s_1, \quad s_2.$$

Not all of these are necessary. For instance, knowing l_1, l_2, θ_1, θ_2 and s_1 should be enough — from these we can find s_2; or we could drop θ_1 and instead use s_1, s_2 and θ_2. Out of the six, **five** are enough to describe the trapezium, and once we know them, everything else follows from trigonometry and we can write down the area.

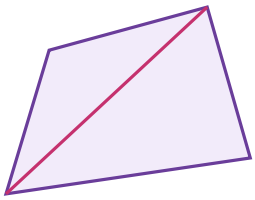
> The main point to emphasise: you keep going back to **figures whose area you already know** — and the basic figure is always the **rectangle**. From rectangles we got parallelograms and triangles; for anything else, split it into rectangles, triangles or parallelograms.

6 Quadrilaterals and polygons: divide and conquer

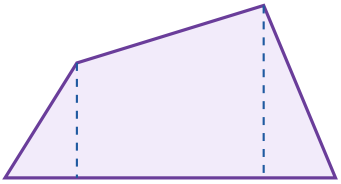
6.1 A general quadrilateral

One option is to draw a **diagonal**, which gives two triangles; with a little work we can compute the area of each, once we know how the quadrilateral is described.

The other option is to **drop perpendiculars**. Then the two end pieces are right-angled triangles, and what remains in the middle is a **trapezium** — which we already know how to handle. And if one of the angles happens to be 90°, we can draw one more line and split that trapezium further into a **rectangle** and another triangle.



two triangles by a diagonal

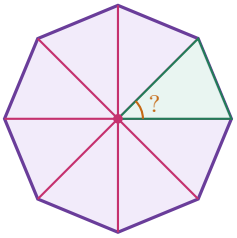


two right triangles + a trapezium

6.2 A regular polygon

For a regular **octagon**, take the centre and draw all the radii. This gives **eight triangles**, and since the octagon is regular they are all **congruent** to one another. So computing the area of *one* triangle is enough:

$$\text{area of octagon} = 8 \times \text{area of one triangle.}$$



one of 8 congruent triangles

How do we compute the area of one such triangle? That is very doable from the given data — you are encouraged to use your trigonometry: work out the angle at the centre, and from it the area. In general, for a regular n-gon the same idea gives the area. And if the polygon is **not** regular, the same idea still works, but each triangle must be computed individually.

In every case the general principle holds: **divide the figure into figures you understand well** — essentially rectangles and triangles — and compute their areas.

7 Non-linear shapes: the circle

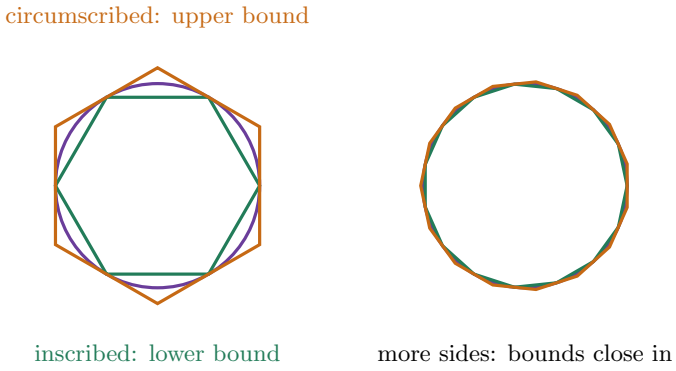
So far every shape has been built from **straight lines**, which is what let us divide it into rectangles and triangles. What if we have a **circle**?

We all know the area of a circle from school, but this was a question that arose in **antiquity**, and one solution was given by **Archimedes**.

7.1 Archimedes' method

Archimedes drew regular polygons **around** the circle — a triangle, then a square, then a heptagon (seven sides), and so on. As the number of sides increases, the polygon approximates the circle better and better; and since we know how to compute the area of a regular n-gon, this gives an **upper bound** for the area of the circle.

He then did the same thing **inside** — a triangle drawn inside, a square inside, a heptagon inside — giving a **lower bound**. Archimedes gave explicit formulae for these polygons, and it is not terribly hard to do.



Then, *in the limit*, the upper bounds and the lower bounds turn out to be the **same** — and therefore the area of the circle must be that common value:

$$\boxed{\text{Area of circle} = \pi r^2}$$

where r is the radius. This is a very beautiful argument, and it was known to Archimedes a very long time ago.

Two things to take away

1. Once again we have reduced everything to **rectangles**: we reduced the circle to polygons, polygons to triangles, and triangles to rectangles.
2. To do this properly — to get the area exactly — the n-gon must be allowed to come really, really close to the circle, which means letting $n \to \infty$. So once again there is a notion of a **limit**.

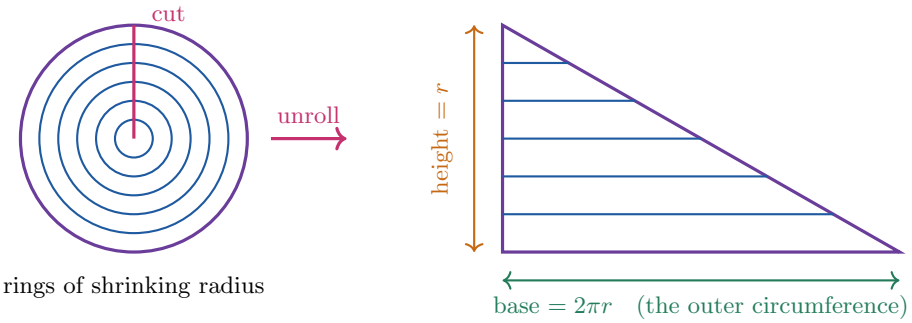
So the slogan is: **divide** into rectangles or figures we understand, and then *gradually* **conquer**. Previously we could divide and conquer directly, because the figures were linear; here we must additionally pass to a limit.

8 A second method: unrolling the circle

Here is another way of doing the same thing, and it is closer to what we shall do next.

Take the circle and break it up by drawing **concentric circles** — circles which are radially shrinking. Now cut along a radius, and **unroll** each ring into a strip. The outermost ring becomes the longest strip, the next one a slightly shorter strip, and so on.

Laying the strips one below another — straightening the bottom edge and the curved edge, while **always preserving the area** — the top edge is deformed, but the whole figure begins to look like a **triangle**.



rings of shrinking radius

By dividing into **smaller and smaller strips** — increasing the number of concentric circles — the strips become thinner and thinner, and the troublesome curvature at the top becomes very, very tiny, so that we can almost ignore it. What is left is a triangle.

- The **height** of that triangle is the length of the radius, that is r.
- The **base** is the length of the longest ring — in other words, the **circumference** of the circle, which is $2\pi r$.

Therefore

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 2\pi r \times r = \boxed{\pi r^2}$$

— exactly as before, obtained again *in the limit*.

> Breaking a shape into **thinner and thinner pieces** and estimating them by **rectangles** is exactly the idea we shall use in the next lecture, when we define the **integral**.

Lecture 4 — Riemann Sums and the Integral

Approximating area by rectangles; left, right, midpoint, upper, lower and trapezoidal sums; the definite integral

1 The strategy from the previous lecture

In the previous lecture we computed areas, and the underlying theme was this: given an arbitrary shape, approximate it as well as you can by **rectangles**, keep dividing into smaller and smaller rectangles, and in the limit the sum of their areas gives the area of the shape. Sometimes we used polygons, trapeziums or triangles instead — but all of those were themselves derived from the rectangle, so the **rectangle was our basic building block**.

A strategy to compute areas

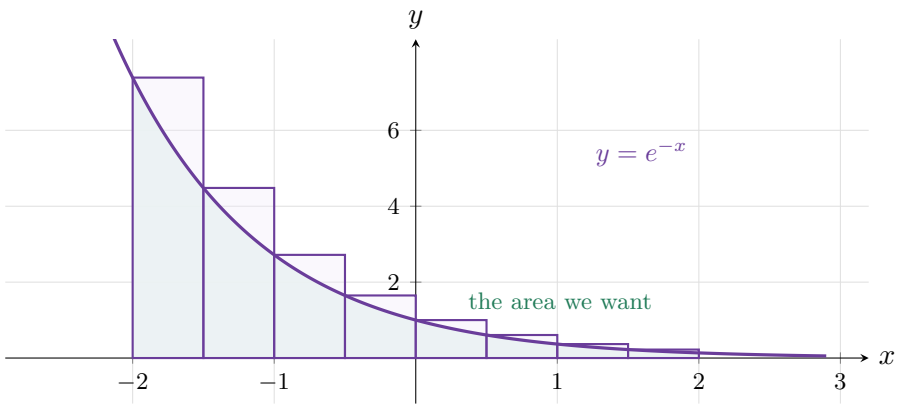
1. Break the shape into **rectangles**. Some part of the shape may be left out, and some extra part may get included.
2. Calculate the area covered by the rectangles.
3. **Decrease the sizes** of the rectangles used. The part left out and the extra part added in both keep decreasing, so the shape is approximated better.
4. Recalculate the area each time. As the sizes decrease, the area is better and better approximated — and **in the limit** we get the area of the shape.

2 The area under a graph

Let us use this strategy in a coherent fashion, and talk about the **area under the graph of a function**. Take the example

$$y = e^{-x}, \qquad \text{between } x = -2 \text{ and } x = 2.$$

In other courses — particularly in **statistics**, which you may have seen already or will see soon — computing exactly this kind of area is important.



We choose a set of rectangles as shown, compute their total area, then **shrink the intervals** and take smaller and smaller rectangles. If the rectangles are small enough, the green area can be

approximated as well as we want; and in the limit — letting the rectangles get smaller and smaller and smaller — we actually compute the area. The same strategy works for *any* graph.

This process was formally introduced in the 1800s, and sums of this kind have a name: they are called **Riemann sums**.

3 Riemann sums, defined

Definition 4.1 — The data P

Let $f : D \to \mathbb{R}$ and suppose the interval $[a, b]$ lies in the domain D. Let P consist of the following data:

- a **partition** of $[a, b]$, that is a choice of intermediate points

$$a = x_0 < x_1 < x_2 < \cdots < x_n = b;$$

- for each small interval $[x_{i-1}, x_i]$, a choice of some number

$$x_i^* \in [x_{i-1},\, x_i].$$

The partition may be chosen **uniform** — the distance from x_0 to x_1 equal to that from x_1 to x_2, and so on — or it may be arbitrary. In our picture above, $x_0 = -2$ and $x_n = 2$, with the intermediate division points in between.

Definition 4.2 — Δx_i and the norm of P

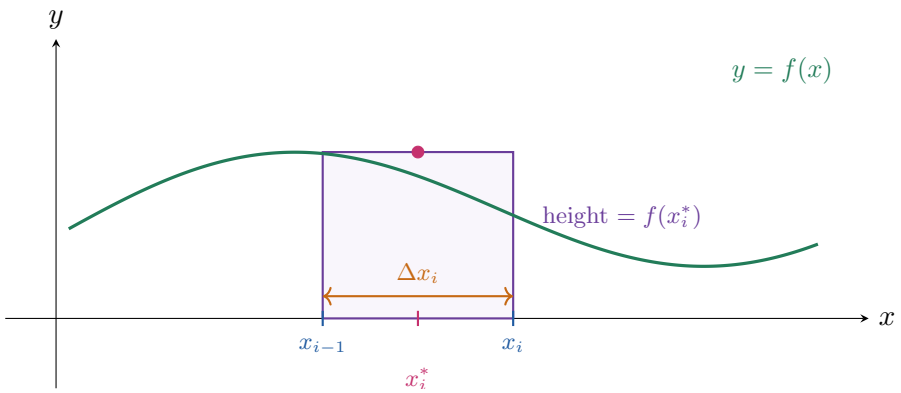
$$\Delta x_i = x_i - x_{i-1} \qquad \text{(the length of the } i\text{-th piece)},$$
$$\|P\| = \max_i \ \Delta x_i \qquad \text{(the } \textbf{norm} \text{ of } P\text{)}.$$

If all the lengths are chosen equal, then $\|P\|$ is simply that common number. In general it is the largest of the piece-lengths.

Definition 4.3 — The Riemann sum

The **Riemann sum** of f with respect to the data P is

$$\boxed{S(P) = \sum_{i=1}^{n} f(x_i^*)\ \Delta x_i}$$



What is going on here? Over the interval $[x_{i-1}, x_i]$ we take the rectangle of height $f(x_i^*)$; multiplying

that height by the length Δx_i gives the area of the rectangle, and summing all of them approximates the area under the graph.

Reading off the first two terms

Suppose we choose $x_i^* = x_{i-1}$, the *left* endpoint of each interval. Then the first term is

$$f(x_0)\,(x_1 - x_0),$$

which is exactly the area of the first rectangle; the second term is

$$f(x_1)\,(x_2 - x_1),$$

the area of the second rectangle; and so on. So with that choice, the Riemann sum *is* the sum of the areas of the rectangles we drew.

Of course, what we would like is to keep **refining** the partition — taking smaller and smaller P — and that will give us something interesting.

4 Different choices of x_i^*

Let us look at a concrete function and compare the various sums. Take

$$f(x) = \sin(2x) - \frac{x^2}{10} + 3, \qquad \text{on the interval } [1, 5],$$

a somewhat complicated-looking function. With $n = 5$ equidistant intervals,

$$\Delta x_i = \frac{5-1}{5} = \frac{4}{5} = 0.8,$$

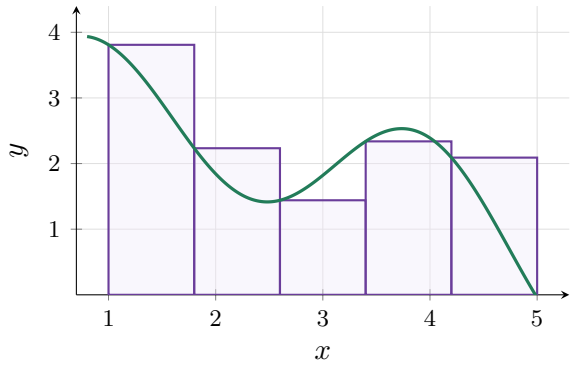
so the intervals are [1, 1.8], [1.8, 2.6], [2.6, 3.4], [3.4, 4.2], [4.2, 5].

Sum	Choice of x_i^*	Behaviour
Left-hand	$x_i^* = x_{i-1}$	9.53 at $n = 5$, then 8.83, closing on the true area
Right-hand	$x_i^* = x_i$	starts at -0.176 for a single rectangle, then 7.42, 7.47, 7.6, …
Midpoint	$x_i^* = \dfrac{x_{i-1} + x_i}{2}$	already 7.28 at the start, then 8.09, 8.08 — a very good approximation
Upper	$f(x_i^*) = \max f$ on the interval	always an **overestimate**; very large initially, improving steadily
Lower	$f(x_i^*) = \min f$ on the interval	always an **underestimate**; starts badly (the minimum is negative), then 6.98, 7.05, …
Trapezoidal	trapeziums instead of rectangles	much better than the others: 8.06, 8.07, 8.07

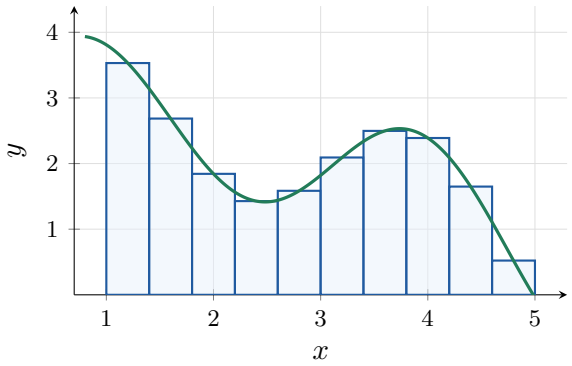
The true area

For this function the true area is ≈ 8.08 square units. Notice how the trapezoidal rule reaches 8.07 almost immediately, while the right-hand sum with a single rectangle gives -0.176 — very far from the actual value.

left-hand sum, $n = 5$



midpoint sum, $n = 10$



As the number of rectangles increases, the intervals shrink and the area under the graph is approximated better and better. (It may not improve at every single step, but in the long run it does.)

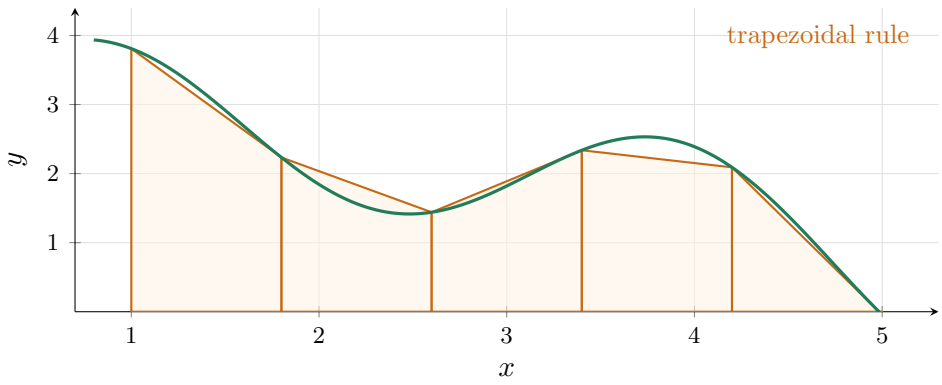
4.1 Upper and lower sums

Instead of choosing a point x_i^*, we may choose the partition and then take the **largest** value of $f(x)$ on $[x_{i-1}, x_i]$ — that gives the **upper sum**, which always overestimates the area. Taking the **smallest** value instead gives the **lower sum**, which always underestimates it.

> This is very similar to the way **Archimedes** obtained the area of a circle: he got overestimates from circumscribed polygons and underestimates from inscribed ones, said the area must lie between the two, and observed that in the limit both give πr^2 — hence the area *is* πr^2. In the same way, the true area lies between the upper sum and the lower sum, and **if in the limit the two agree, that common value is the area**.

4.2 The trapezoidal rule

One last variation: instead of rectangles, use **trapeziums**. Over $[x_{i-1}, x_i]$ we join $f(x_{i-1})$ to $f(x_i)$ by a straight line, giving a trapezium with a right angle at the base. This is the **trapezoidal rule**, which you may have met in school, and it is a much better approximation than the others.



An idea to keep in mind

Instead of joining the two endpoints by a chord, we could take a line *slightly above* — namely the **tangent**. That would also give an approximation, and something interesting comes out of

it. Keep this idea in mind; it comes up in the next lecture.

So we shall use Riemann sums based on rectangles, but other kinds are possible — and if the function is what is called **Riemann integrable**, it does not really matter which kind we take: in the limit they all give the same thing, namely the area.

5 The definite integral

Definition 4.4 — The definite integral

Let $f : D \to \mathbb{R}$ and suppose $[a, b] \subseteq D$. The **definite integral of f from a to b** is

$$\lim_{\|P\|\to 0} S(P) = \lim_{\|P\|\to 0} \sum_{i=1}^{n} f(x_i^*)\ \Delta x_i \ = \int_a^b f(x)\,dx$$

We have already seen this idea in the animations, where the number of intervals increased — in other words the interval sizes decreased — and the interval sizes are exactly what $\|P\|$ keeps track of.

Two remarks on the notation

- It is called the *definite* integral because there is something else called the **indefinite** integral, which we shall see in a later lecture. If we say "from a to b" we need not use the word definite; if we do not say it, we had better say *definite* to indicate that it is over some interval.
- The dx is **very crucial**. If you drop the dx you may run into trouble later on.

5.1 When does the integral give the area?

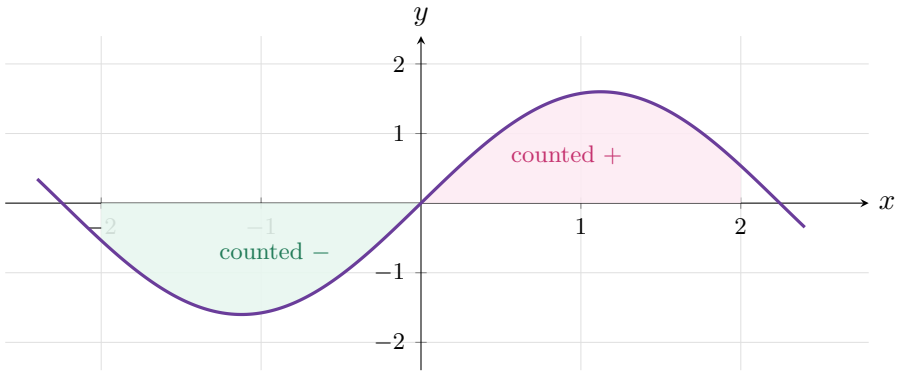
Integral and area

If f is **positive** and **piecewise continuous** on $[a, b]$ — continuity being allowed to fail at finitely many points — then the area under the graph of f above the interval $[a, b]$ is measured by

$$\int_a^b f(x)\,dx.$$

And if f is **negative**? Note that the definition never assumed anything about the sign — it holds for any function. If f is negative, the number still measures an area, namely the area *above* the graph and *below* the interval $[a, b]$; but since each $f(x_i^*)$ is negative, the sum comes out negative. So it must be qualified with a minus sign: it is $-\int_a^b f(x)\,dx$ that is the area.

In other words, what the integral computes is **signed area**.



So on an interval where the graph goes above and below the axis, the integral computes the area above *minus* the area below. Keep that in mind whenever you use an integral to compute area.

6 A worked example from first principles

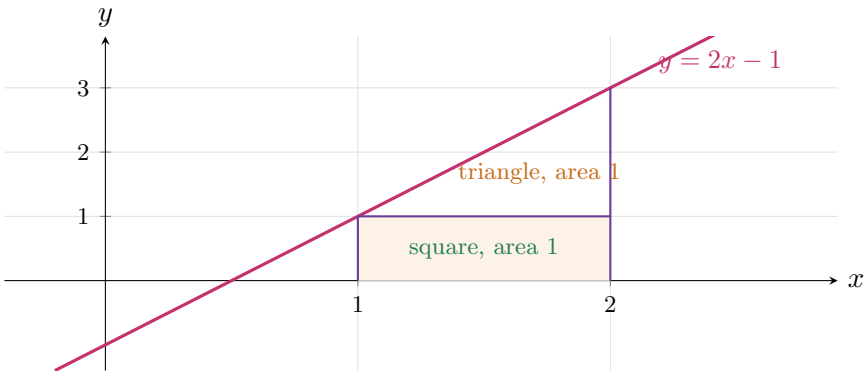
Example 4.1

Compute the area under $f(x) = 2x - 1$ between $x = 1$ and $x = 2$.

By inspection

This is the simplest area we could ask for, and we can read it off the figure. The region splits into a **square** of side 1, and a **triangle** with base 1 and height 2:

$$\text{area} = \underbrace{1}_{\text{square}} + \underbrace{\tfrac{1}{2} \times 1 \times 2}_{\text{triangle}} = 1 + 1 = \boxed{2 \text{ square units}}$$



By Riemann sums

Now let us do the same thing from **first principles**. Take the partition P_n dividing $[1, 2]$ into n equal parts:

$$x_i = 1 + \frac{i}{n}, \qquad \text{so} \qquad 1 = x_0 < x_1 < x_2 < \cdots < x_n = 1 + \frac{n}{n} = 2.$$

Each part has size

$$\Delta x_i = \frac{1}{n}, \qquad \text{hence} \qquad \|P_n\| = \frac{1}{n},$$

and letting $n \to \infty$ makes $\|P_n\| \to 0$.

Choose the **right-hand** sum, that is $x_i^* = x_i = 1 + \dfrac{i}{n}$. Then

$$S(P_n) = \sum_{i=1}^{n} f(x_i^*)\,\Delta x_i = \sum_{i=1}^{n} \left[2\left(1+\frac{i}{n}\right) - 1\right]\frac{1}{n}$$

$$= \sum_{i=1}^{n} \left(1 + \frac{2i}{n}\right)\frac{1}{n} = \sum_{i=1}^{n} \frac{1}{n} \; + \sum_{i=1}^{n} \frac{2i}{n^2}.$$

First sum. Here $\frac{1}{n}$ is a constant, so it comes out and we are left with a sum of n ones:

$$\sum_{i=1}^{n} \frac{1}{n} = \frac{n}{n} = 1.$$

Second sum. Here $\frac{2}{n^2}$ is a constant and what remains is $\sum_{i=1}^{n} i$, which we know:

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2} \qquad \Longrightarrow \qquad \frac{2}{n^2}\cdot\frac{n(n+1)}{2} = \frac{n+1}{n} = 1 + \frac{1}{n}.$$

Adding the two,

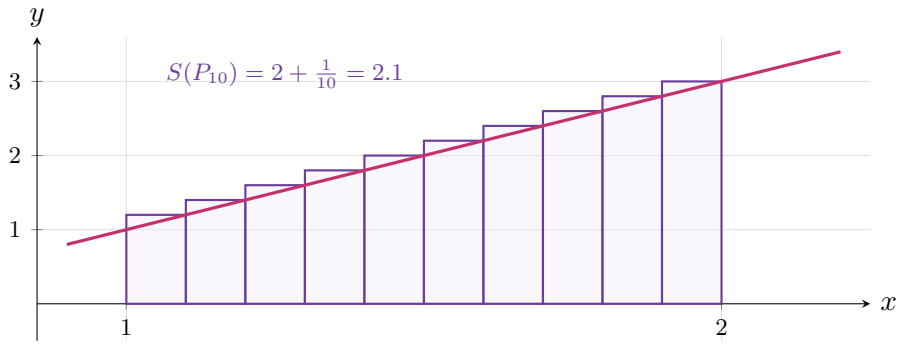
$$S(P_n) = 1 + 1 + \frac{1}{n} = 2 + \frac{1}{n}.$$

Finally, letting $n \to \infty$ — which is the same as letting $\|P_n\| \to 0$ —

$$\int_1^2 (2x-1)\,dx = \lim_{n\to\infty}\left(2 + \frac{1}{n}\right) = \boxed{2}$$

exactly the 2 square units we obtained by observation.

right-hand Riemann sum, $n = 10$



7 Why we need something better

Notice the main problem: even for a **linear** function, this was a fairly non-trivial computation to carry out. For a harder function — say e^{-x}, or a polynomial such as $x^3 - 2x^2 + 5$ — using this method directly would be really difficult. It *can* be done, but it is not an easy process.

So we need a way to compute the integral **more efficiently** — and that is what we shall see in the next lecture.

Lecture 5 — Integrals as Anti-derivatives

The indefinite integral; the fundamental theorem of calculus; a table of integrals; worked examples

1 Recalling the definite integral

In the previous lecture we defined integrals via **Riemann sums**. Let us recall that. For f defined on a domain containing $[a, b]$, we took a partition

$$a = x_0 < x_1 < \cdots < x_n = b,$$

chose a point $x_i^* \in [x_{i-1}, x_i]$ in each sub-interval, and set

$$\Delta x_i = x_i - x_{i-1}, \qquad \|P\| = \max_i \Delta x_i, \qquad S(P) = \sum_{i=1}^{n} f(x_i^*)\,\Delta x_i.$$

The rectangles had base Δx_i and height $f(x_i^*)$, and their total area approximated the area above $[x_{i-1}, x_i]$ and below the graph of f — with f positive; if f were negative the same thing would give the area *above* the graph, computed with a negative sign. Making the partitions finer and finer gave

$$\int_a^b f(x)\,dx = \lim_{\|P\|\to 0} S(P) = \lim_{\|P\|\to 0} \sum_{i=1}^{n} f(x_i^*)\,\Delta x_i.$$

Reading the notation

The leftmost sign is the **integral** sign; the smaller number a is written at the bottom, the larger number b at the top; then comes the function to be integrated; and then dx, which among other things indicates the **end** of the integral we want to compute.

We saw that integrating $2x - 1$ from 1 to 2 was already a fairly non-trivial computation. For more complicated functions we would presumably have to do much more work — and that rather defeats the purpose of defining the integral, which is supposed to compute areas *quickly*.

So in this lecture we link the integral with something we studied even earlier — the **derivative**, or rather the **anti-derivative**. Once that link is understood, computing integrals becomes a very routine process.

2 Anti-derivatives

Definition 5.1 — Anti-derivative

Let $f : D \to \mathbb{R}$, with $D \subseteq \mathbb{R}$. An **anti-derivative** of f is a function F defined on D such that

$$\boxed{F'(x) = f(x) \qquad \text{for all } x \in D.}$$

In particular this assumes F is differentiable, and that on computing its derivative we get $f(x)$. So the anti-derivative simply **reverses the operation of differentiation**: we ask whether there is a function F which, when differentiated, gives f.

Example 5.1

Let

$$f(x) = x^7 + 2x^6 - \pi x^5 + 0.5\,x^4 - 9.$$

Then an anti-derivative is

$$F(x) = \frac{x^8}{8} + \frac{2x^7}{7} - \frac{\pi x^6}{6} + \frac{0.5\,x^5}{5} - 9x.$$

If you are wondering how this came about: take this $F(x)$, differentiate it, and check that you get $f(x)$. We know how to differentiate, so we simply try to go the other way and ask how to "anti-differentiate".

2.1 Adding a constant

Fact

If F is an anti-derivative of f and c is any real number, then

$$F_1(x) = F(x) + c$$

is **also** an anti-derivative of f.

Proof

Differentiate F_1. By linearity,

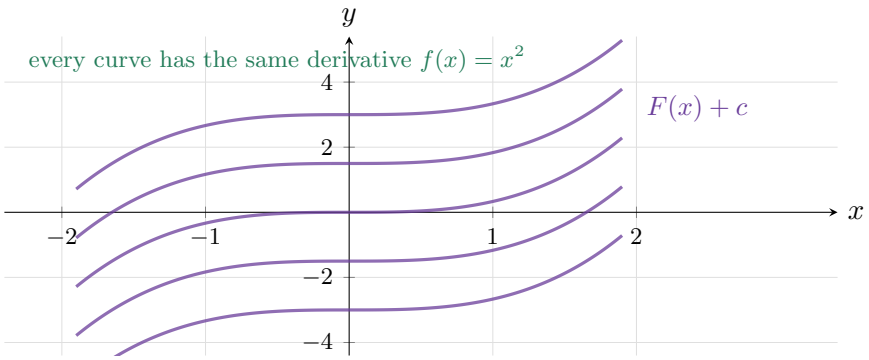
$$F_1'(x) = F'(x) + \frac{d}{dx}(c) = f(x) + 0 = f(x),$$

since the derivative of a constant is 0. ■

This much is quite obvious. What is far more **remarkable** is the converse.

> **Every** anti-derivative has this form. So once you identify *one* anti-derivative of a function, you know *all* possible anti-derivatives.

For the example above, adding 5 to $F(x)$ gives another anti-derivative; so does adding -3000. And any other anti-derivative whatsoever must be $F(x)$ plus some constant — there is no other possibility.



Because of this we say **the** anti-derivative rather than *an* anti-derivative: even though it is not unique, the only possible difference between two anti-derivatives is a constant — and we shall see that constants do not contribute to what we want the anti-derivative for. Accordingly, we always write the constant explicitly:

$$F(x) = \frac{x^8}{8} + \frac{2x^7}{7} - \frac{\pi x^6}{6} + \frac{0.5\,x^5}{5} - 9x \boxed{+\,c}, \qquad c \in \mathbb{R}.$$

3 The indefinite integral

Definition 5.2 — Indefinite integral

The anti-derivative of a function f is more often called the **indefinite integral** of f, and is denoted

$$\int f(x)\,dx.$$

Note that here there is **no** a **and no** b — there are no limits of integration.

Definite integral $\int_a^b f(x)\,dx$	**Indefinite integral** $\int f(x)\,dx$
Is a **number**	Is a **function**
Depends on the limits a and b — change them and you get a different number	Has no limits; it is simply the anti-derivative

When we say "the integral of f" we mean this function — the anti-derivative.

4 Do they exist?

Two questions are worth pausing over: one left over from the previous lecture, one from this.

Existence

- The definite integral is defined as a **limit**, and limits may not always exist. So the definite integral **may not exist** — there are plenty of examples. **However**, if f is a **continuous** function on $[a, b]$, then the definite integral *does* exist.
- Likewise, anti-derivatives **may not always exist**. But once again, if the function is nice — in particular if it is **continuous** — then they do exist.

So far, then: the **definite integral** from a to b is a number, and the idea is that it computes the area below or above the graph of f between the axes — keeping track of the fact that the area comes with a minus sign where the function is below the axis, and a plus sign where it is above. The **indefinite integral** is nothing but an anti-derivative: the function which, on being differentiated, gives $f(x)$.

Why then do we call it the *integral*? Because there is a connection between the two, and that connection is explained by the **fundamental theorem of calculus**.

5 The fundamental theorem of calculus

About the name

The theorem is typically attributed to **Newton** — although it was computationally known before him, Newton formulated it as a theorem, possibly for the first time, and that is why he is often credited as one of the inventors of calculus. And when we say something is *fundamental*, it really means that it gives a very deep insight into the subject — which is exactly what this theorem does.

Fundamental theorem of calculus — first form

Suppose f is **continuous** on a domain D which includes the interval $[a, b]$. Then an anti-derivative for f on $[a, b]$ is given by

$$\boxed{F(x) = \int_a^x f(t)\,dt}$$

Why the variable inside is called t

Since we are writing $F(x)$, our variable is x — and x appears in the integral as the **upper limit**. We cannot then use x *inside* the integral as well, so we integrate with respect to some other variable, which we call t. Do not get confused: this is the same function f; only the variable has been given a different name.

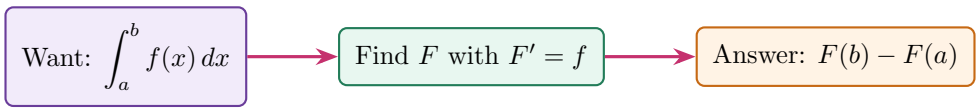
So the first form says two things: the anti-derivative **exists**, and it is in fact given by that definite integral. This is of course a *theoretical* statement, since the definite integral was itself defined via a limit and is rather hard to compute. What is really useful — and what we shall use soon — is the second form.

Fundamental theorem of calculus — second form

Suppose f is continuous on a domain D including $[a, b]$, and let F be the indefinite integral of f (that is, differentiating F gives f). Then

$$\boxed{\int_a^b f(x)\,dx = F(b) - F(a)}$$

This is telling us something interesting: to compute the definite integral from a to b, **find an anti-derivative** — the indefinite integral — then evaluate it at b and subtract the same thing evaluated at a. This second form is of immense use, and it is what we shall use to compute integrals.



6 A table of derivatives and integrals

Why a table of derivatives *and* integrals? Because the fundamental theorem tells us that if we want a definite integral, we had better know the **indefinite** integral. And how do we know these integrals? By **back-calculating**: we know how derivatives work, and from there we work out how integrals work.

Derivatives and integrals

$f(x)$	$f'(x)$	$\int f(x)\,dx$
1	0	x
x^r	$r\,x^{r-1}$	$\dfrac{x^{r+1}}{r+1}$ $(r \neq -1)$; $\log\lvert x\rvert$ $(r=-1)$
$\sin x$	$\cos x$	$-\cos x$
$\cos x$	$-\sin x$	$\sin x$
$\tan x$	$\sec^2 x$	$\log\lvert\sec x\rvert$
$\sec x$	$\sec x \tan x$	$\log\lvert\sec x + \tan x\rvert$
$\cot x$	$-\operatorname{cosec}^2 x$	$\log\lvert\sin x\rvert$
$\operatorname{cosec} x$	$-\operatorname{cosec} x\,\cot x$	$\log\lvert\operatorname{cosec} x - \cot x\rvert$
$e^{\lambda x}$	$\lambda\, e^{\lambda x}$	$\dfrac{e^{\lambda x}}{\lambda}$
$a^x \quad (a>0)$	$a^x \log a$	$\dfrac{a^x}{\log a}$
$\log x$	$\dfrac{1}{x}$	$x\log x - x$
$\dfrac{1}{\sqrt{a^2-x^2}}$	—	$\sin^{-1}\left(\dfrac{x}{a}\right)$
$\dfrac{1}{a^2+x^2}$	—	$\dfrac{1}{a}\tan^{-1}\left(\dfrac{x}{a}\right)$
$\dfrac{1}{a^2-x^2}$	—	$\dfrac{1}{2a}\log\left\lvert\dfrac{x+a}{x-a}\right\rvert$

Reading a few rows:

- For the function 1, the derivative is 0 and the integral is x — because differentiating x gives 1.
- For x^r the integral depends on r: it is $\dfrac{x^{r+1}}{r+1}$ provided $r \neq -1$. If $r = -1$ the denominator becomes 0, and indeed for $r = -1$ we do not get a polynomial at all: the function is $\frac{1}{x}$ and its integral is $\log|x|$ — something we saw when doing derivatives.
- The last four trigonometric rows can be worked out from the earlier ones.
- In the final three rows the derivatives are omitted only because they are tedious calculations; you can do them yourself using the quotient rule together with the rule for compositions.

How to use the table

The way to check any row is to take the function in the **rightmost** column, **differentiate it**, and see that you get the function in the **leftmost** column. Most of these anti-derivatives are in fact obtained precisely by such observation, and you are encouraged to verify each one by hand.

7 Worked examples

Example 5.2

Compute $\displaystyle\int_1^2 x\,dx$.

Apply the fundamental theorem: find the indefinite integral, evaluate it at 2, evaluate it at 1, and subtract. The anti-derivative of x is $\frac{x^2}{2}$ — by observation, or from the table. Rather than writing those as separate steps, the standard notation is a square bracket with the limits at the right:

$$\int_1^2 x\,dx = \left[\frac{x^2}{2}\right]_1^2 = \frac{2^2}{2} - \frac{1^2}{2} = \frac{4}{2} - \frac{1}{2} = \boxed{\frac{3}{2}}$$

Example 5.3

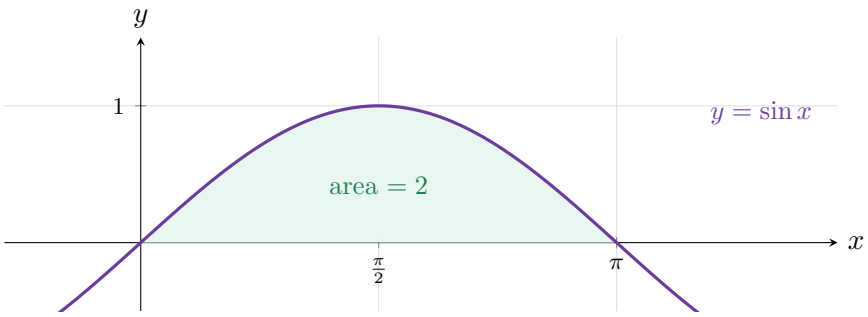
Compute $\displaystyle\int_0^\pi \sin x\,dx$.

The anti-derivative of $\sin x$ is $-\cos x$. Here there is a minus sign floating around, so it is best to use plenty of brackets to avoid mistakes:

$$\int_0^\pi \sin x\,dx = \Big[-\cos x\Big]_0^\pi = \big(-\cos\pi\big) - \big(-\cos 0\big).$$

Now $\cos\pi = -1$, so the first term is $-(-1) = 1$; and $\cos 0 = 1$, so the second term is $-(-1) = +1$ — a minus and a minus giving a plus. Hence

$$\int_0^\pi \sin x\,dx = 1 + 1 = \boxed{2}$$



So this tells us that if we take the curve $y = \sin x$ and ask for the area of one arch, between 0 and π, that area is exactly 2. Here you can see the **power of the integral**: this area is not at all easy to compute otherwise — we would have had no real way of guessing it, and would have had to go through Riemann sums, obtain a rather difficult sum, and evaluate the limit. The fundamental theorem tells us in one shot that the area is 2.

8 An improper integral

Example 5.4

Compute $\displaystyle\int_0^\infty e^{-2x}\,dx$.

There is a very important reason for doing this example: you will see functions like this in other courses — for instance in **statistics**, where this **exponential decay** appears as what is called a **density function**. Note also our notation: the function is little f and its anti-derivative is capital F, which may connect with what you meet there.

What $\int_a^\infty$ means

We have only defined the definite integral between two numbers a and b. For an infinite upper limit we define

$$\int_a^\infty f(x)\,dx \;=\; \lim_{b\to\infty}\int_a^b f(x)\,dx.$$

Several things are involved in this definition: the integral from a to b must itself be defined, *and* the limit of those numbers as $b \to \infty$ must exist. So it is a fairly non-trivial definition — the good part is that we compute it in the same way.

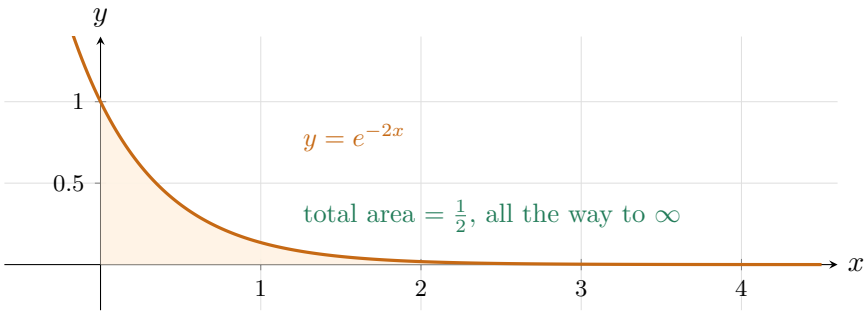
From the table, an anti-derivative of $e^{\lambda x}$ is $\dfrac{e^{\lambda x}}{\lambda}$, so with $\lambda = -2$:

$$\int_0^\infty e^{-2x}\,dx = \lim_{b\to\infty}\int_0^b e^{-2x}\,dx = \lim_{b\to\infty}\left[\frac{e^{-2x}}{-2}\right]_0^b$$

$$= \lim_{b\to\infty}\left(-\frac{1}{2}\right)\left(e^{-2b}-e^{0}\right) = \lim_{b\to\infty}\left(\frac{1}{2}-\frac{e^{-2b}}{2}\right).$$

And as $b \to \infty$ we know that $e^{-2b} \to 0$. Therefore

$$\int_0^\infty e^{-2x}\,dx = \boxed{\frac{1}{2}}$$



A connection with statistics

If instead of e^{-2x} we had taken $2\,e^{-2x}$, the integral would have been **1**. That should ring a bell from your statistics course — it is the **exponential random variable**, whose density must integrate to 1.

9 Summary of the method

- The **anti-derivative** — also called the **integral**, or the **indefinite integral** — of a function f exists when f is continuous.
- We find it using our knowledge of **derivatives**: knowing how to differentiate functions, we manipulate and work out what the integrals are.
- The key point: we use these indefinite integrals to compute **definite integrals**. This is the main thrust of the **fundamental theorem of calculus** — to compute $\int_a^b f(x)\,dx$, find the indefinite integral F, evaluate it at b, evaluate it at a, and take the difference.

In the next lecture we shall see more examples of computing definite integrals, together with some rules for computing them. Since all of this is connected with differentiation, it should come as no surprise that the same kinds of properties we had for derivatives have counterparts for integrals — and we will use those in computing more integrals.

Lecture 6 — Computing Integrals and Areas

Properties of integrals; integration by parts; piecewise functions; substitution; the area of a circle

1 Where we stand

We have now reached a place where we know how to compute a **definite integral**, and we know that a definite integral represents area — what is colloquially called the *area under the curve* (with the curve possibly below the x-axis, which we interpret suitably). Now we use this to **compute areas**.

Recall the two things we have:

The definite integral was defined as the limit of Riemann sums as the underlying partitions become finer and finer. But this was a rather difficult approach, because **computing limits is hard**. So we always try to bypass limits — by making good hypotheses on the functions, or by proving theorems that let us compute them in other ways.

Newton's theorem did exactly that: it let us compute integrals via the indefinite integral, that is the anti-derivative, linking differentiation and integration. If f is continuous on a domain D including $[a, b]$, and F is the integral of f (so $F' = f$), then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

So the upshot: if one knows the indefinite integral of a continuous function, one can use it to compute the area between the graph of the function and an interval on the x-axis.

2 Basic properties of indefinite integrals

These are on the lines of the properties for derivatives, and the proofs are essentially the same — they follow from the corresponding properties of the derivative. You are strongly encouraged to prove them yourself; we prove only the last one here.

Properties of the indefinite integral

$$\int c\,f(x)\,dx = c\int f(x)\,dx, \qquad c \in \mathbb{R} \text{ a constant}$$

$$\int \big(f+g\big)(x)\,dx = \int f(x)\,dx + \int g(x)\,dx$$

$$\int \big(f-g\big)(x)\,dx = \int f(x)\,dx - \int g(x)\,dx$$

$$\int \big(f\,g'\big)(x)\,dx = (fg)(x) - \int \big(f'g\big)(x)\,dx \qquad \textbf{(integration by parts)}$$

2.1 Proof of integration by parts

Start from fg and ask what its derivative is. By the **product rule**,

$$\frac{d}{dx}\big(fg\big)(x) = f'(x)\,g(x) + g'(x)\,f(x) = \big(f'g\big)(x) + \big(g'f\big)(x).$$

Since the derivative of fg is the function on the right, the **integral** of the function on the right is fg:

$$\int \Big[\big(f'g\big) + \big(g'f\big)\Big](x)\,dx = (fg)(x).$$

Now this is a sum, so by the second property we may split it:

$$\int \big(f'g\big)(x)\,dx \;+\; \int \big(g'f\big)(x)\,dx = (fg)(x).$$

Moving the first term to the other side gives exactly what we want:

$$\boxed{\int \big(g'f\big)(x)\,dx = (fg)(x) - \int \big(f'g\big)(x)\,dx}$$

■

This is called **integration by parts**, and although it is an easy trick, it is very useful in practice.

3 Properties of definite integrals

The first few follow from the properties of indefinite integrals — either from first principles, or via Newton's theorem by writing both sides in terms of F.

Properties of the definite integral

$$\int_a^b c\,f(x)\,dx = c\int_a^b f(x)\,dx$$

$$\int_a^b \big(f \pm g\big)(x)\,dx = \int_a^b f(x)\,dx \pm \int_a^b g(x)\,dx$$

$$\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx$$

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx, \qquad c \in \mathbb{R}$$

$$f(x) \geq g(x) \text{ on } [a,b] \quad \Longrightarrow \quad \int_a^b f(x)\,dx \geq \int_a^b g(x)\,dx$$

3.1 Reading the reversal rule

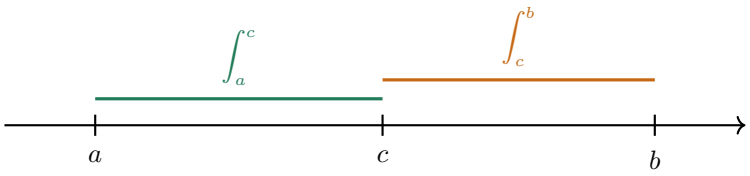
The rule $\int_b^a = -\int_a^b$ may look strange at first. There are two ways to interpret it:

- Interpret b to a as going **backwards** along the interval. Then in the Riemann sums the Δx_i would be **negative** lengths, so the sums become the negative of the usual ones, and we pick up the minus sign.

- Or simply take this as a **definition**. Either way, the statement is very natural: the right-hand side we already understand, and the left-hand side is either defined this way, or the definition of the definite integral is broadened to allow the starting point to be larger than the ending point.

3.2 Reading the splitting rule

If c lies **between** a and b, splitting the interval seems perfectly natural. If c is *not* between them — say $c < a$, or $c > b$ — the statement must be interpreted in the light of the reversal rule above: the first piece subtracts out what the second piece adds back.



3.3 Two consequences of the inequality

- If $f(x) = g(x)$ for all **but finitely many** points of $[a, b]$, then the two integrals are **equal**. The take-home message: the function **need not be continuous** everywhere. Even if f fails to be continuous at a few points, we may treat it as if it were and compute the integral in the same way.
- Taking $g(x) = 0$: if $f(x) \geq 0$ then $\displaystyle\int_a^b f(x)\,dx \geq 0$ — natural, since in that case the integral is computing an area.

4 Example 1: a polynomial

Example 6.1

Compute $\displaystyle\int_1^3 \left(x^2 - 4x + 2\right) dx.$

Break it up using linearity:

$$\int_1^3 \left(x^2 - 4x + 2\right) dx = \int_1^3 x^2\,dx \;-\; 4\int_1^3 x\,dx \;+\; 2\int_1^3 1\,dx.$$

We know the anti-derivative of each piece, using $\displaystyle\int x^n\,dx = \frac{x^{n+1}}{n+1}$:

$$= \left[\frac{x^3}{3}\right]_1^3 - 4\left[\frac{x^2}{2}\right]_1^3 + 2\Big[x\Big]_1^3.$$

Substituting the limits,

$$= \frac{1}{3}\left(3^3 - 1^3\right) - \frac{4}{2}\left(3^2 - 1^2\right) + 2\left(3 - 1\right) = \frac{26}{3} - 2(8) + 2(2) = \frac{26}{3} - 16 + 4 = \frac{26}{3} - 12.$$

Hence

$$\boxed{\int_1^3 \left(x^2 - 4x + 2\right) dx = \frac{26 - 36}{3} = -\frac{10}{3}}$$

A negative answer

The value is negative because the parabola lies *below* the x-axis throughout $[1,3]$ — its roots are $2 \pm \sqrt{2} \approx 0.59$ and 3.41, so the whole interval sits between them. The integral is computing the signed area, as we discussed earlier.

5 Example 2: using symmetry

Example 6.2

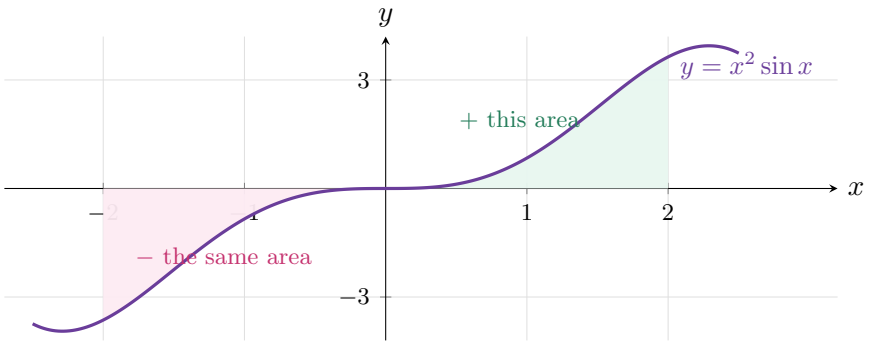
Compute $\displaystyle\int_{-2}^{2} x^2 \sin x \, dx$.

There is no obvious way of calculating this from what we have done so far: we would need the anti-derivative of $x^2 \sin x$, which can be found by integration by parts but requires some work. Instead the trick is to **observe the limits of integration** and **what kind of function** $x^2 \sin x$ is.

Put $f(x) = x^2 \sin x$. Then

$$f(-x) = (-x)^2 \sin(-x) = x^2 \, (-\sin x) = -\, x^2 \sin x = -f(x),$$

since $(-x)^2 = x^2$ and $\sin(-x) = -\sin x$. So, for instance, $f(-1) = -f(1)$ and $f(-2) = -f(2)$.



Split the integral at 0, which one of our rules allows:

$$\int_{-2}^{2} x^2 \sin x \, dx = \int_{-2}^{0} x^2 \sin x \, dx + \int_{0}^{2} x^2 \sin x \, dx.$$

The first part computes that area but with a **negative** sign; the second computes the mirror area with a **positive** sign. The two areas are clearly the same, so

$$\int_{-2}^{0} x^2 \sin x \, dx = -\int_{0}^{2} x^2 \sin x \, dx,$$

and adding them gives

$$\boxed{\int_{-2}^{2} x^2 \sin x \, dx = 0}$$

This works for **any** function satisfying $f(-x) = -f(x)$, whenever the limits of integration are **symmetric about** 0.

6 Integration of piecewise defined functions

If f is defined piecewise on sub-intervals of $[a, b]$, then the definite integral from a to b is computed by integrating on each sub-interval and **adding them up**.

Example 6.3

Let

$$f(x) = \begin{cases} x, & 0 \le x \le 1, \\ 3 - x, & 1 < x \le 2. \end{cases}$$

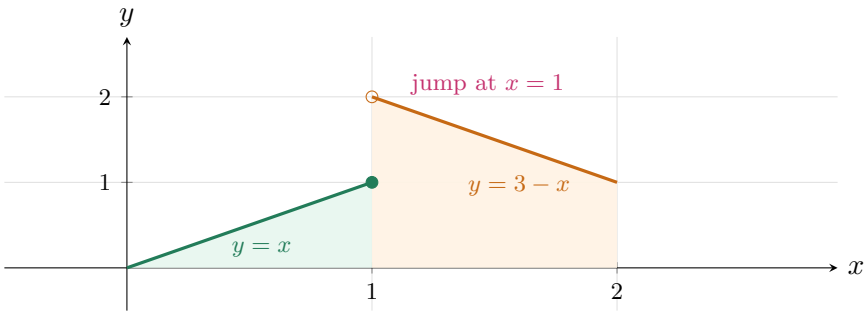
Compute $\displaystyle\int_0^2 f(x)\,dx$.

Break it into the parts on which f is defined:

$$\int_0^2 f(x)\,dx = \int_0^1 f(x)\,dx + \int_1^2 f(x)\,dx = \int_0^1 x\,dx + \int_1^2 (3 - x)\,dx.$$

Not continuous — and that is fine

Notice that this function is **not continuous** at $x = 1$: the left limit is 1, whereas the right limit is $3 - 1 = 2$. But that is all right — if a function fails to be continuous at a few points, we can still do our integrals with no problem.



Computing each piece:

$$\int_0^1 x\,dx = \left[\frac{x^2}{2}\right]_0^1 = \frac{1}{2}, \qquad \int_1^2 (3 - x)\,dx = \left[3x - \frac{x^2}{2}\right]_1^2 = (6 - 2) - \left(3 - \frac{1}{2}\right) = 4 - \frac{5}{2} = \frac{3}{2}.$$

Adding,

$$\boxed{\int_0^2 f(x)\,dx = \frac{1}{2} + \frac{3}{2} = 2}$$

7 Integration by parts for definite integrals

Applying the rule between limits:

$$\boxed{\int_a^b (fg')(x)\,dx = \Big[(fg)(x)\Big]_a^b - \int_a^b (f'g)(x)\,dx}$$

that is, take fg at b minus fg at a, and then subtract the remaining integral.

Example 6.4

Compute $\displaystyle\int_0^{\infty} 3x\, e^{-3x}\, dx.$

(The infinite limit is deliberate — to make you comfortable with integrating up to infinity. You may well meet this integral again in your statistics course.) We integrate from 0 to b and then let $b \to \infty$.

Choosing f and g

The key question is what to take as f and what as g. The idea is to choose f so that f' becomes **nice**, making the last integral easy to compute. So take

$$f(x) = x \qquad \Longrightarrow \qquad f'(x) = 1 \quad \text{(very nice)},$$
$$g'(x) = 3e^{-3x} \qquad \Longrightarrow \qquad g(x) = -e^{-3x}.$$

Check the last one: differentiating $-e^{-3x}$ gives $-(-3)e^{-3x} = 3e^{-3x}$, so indeed $-e^{-3x}$ is the integral of $3e^{-3x}$.

Applying the formula

$$\begin{aligned}\int_0^b 3x\, e^{-3x}\, dx &= \Big[\, x \cdot \big(-e^{-3x}\big) \,\Big]_0^b - \int_0^b 1 \cdot \big(-e^{-3x}\big)\, dx \\ &= -b\, e^{-3b} + \int_0^b e^{-3x}\, dx \\ &= -b\, e^{-3b} + \left[\frac{-e^{-3x}}{3}\right]_0^b = -b\, e^{-3b} - \frac{e^{-3b}}{3} + \frac{1}{3}.\end{aligned}$$

Taking the limit

Now let $b \to \infty$. The last term is a constant and stays. The second term $\frac{e^{-3b}}{3}$ goes directly to 0. The first term has e^{-3b} multiplied by b — but recall from our discussion of how fast functions grow:

The exponential beats the polynomial.

So e^{-3b} goes to 0 much faster than b increases, and $b\, e^{-3b} \to 0$ as well. Therefore

$$\boxed{\int_0^{\infty} 3x\, e^{-3x}\, dx = \frac{1}{3}}$$

Why bother with the technique?

There are other ways of solving this — you could be clever and guess an anti-derivative (which is essentially what integration by parts just produced). But it slowly becomes difficult to always be clever. Instead, apply these nice results and keep **simplifying** the problem until you reduce it to integrals you know. That is the idea behind all these techniques.

8 Substitution

The substitution rule

$$\int_a^b f\big(g(x)\big)\, g'(x)\, dx = \int_{g(a)}^{g(b)} f(u)\, du$$

where u plays the role of $g(x)$.

Although it is written from left to right, in practice one often goes **right to left**: take u and substitute $u = g(x)$.

Example 6.5

Compute $\displaystyle\int_0^a \sqrt{a^2 - u^2}\, du$.

The substitution and the new limits

Put

$$u = g(x) = a \sin x.$$

The limits for u run from 0 to a; correspondingly, since

$$g(0) = a \sin 0 = 0 \qquad \text{and} \qquad g\big(\tfrac{\pi}{2}\big) = a \sin \tfrac{\pi}{2} = a,$$

the limits for x run from 0 to $\frac{\pi}{2}$.

Here $f(u) = \sqrt{a^2 - u^2}$, so

$$f\big(g(x)\big) = \sqrt{a^2 - \big(a \sin x\big)^2} = \sqrt{a^2 - a^2 \sin^2 x},$$

and $g'(x) = a \cos x$. So the integral becomes

$$\int_0^a \sqrt{a^2 - u^2}\, du = \int_0^{\pi/2} \sqrt{a^2 - a^2 \sin^2 x}\ \ a \cos x\, dx.$$

Simplifying

Take a common out of the square root and use $\sin^2 x + \cos^2 x = 1$:

$$\sqrt{a^2 - a^2 \sin^2 x} = a\sqrt{1 - \sin^2 x} = a\sqrt{\cos^2 x} = a \cos x,$$

where the last step uses $\sqrt{\cos^2 x} = |\cos x| = \cos x$, valid because $\cos x \geq 0$ on $\big[0, \frac{\pi}{2}\big]$. So the integral is

$$\int_0^{\pi/2} a \cos x \cdot a \cos x\, dx = a^2 \int_0^{\pi/2} \cos^2 x\, dx.$$

Now express $\cos^2 x$ in terms of $\cos 2x$ — the same idea we used earlier:

$$\cos^2 x = \frac{1 + \cos 2x}{2} \qquad \Longrightarrow \qquad a^2 \int_0^{\pi/2} \frac{1 + \cos 2x}{2}\, dx = \frac{a^2}{2}\left[x + \frac{\sin 2x}{2}\right]_0^{\pi/2},$$

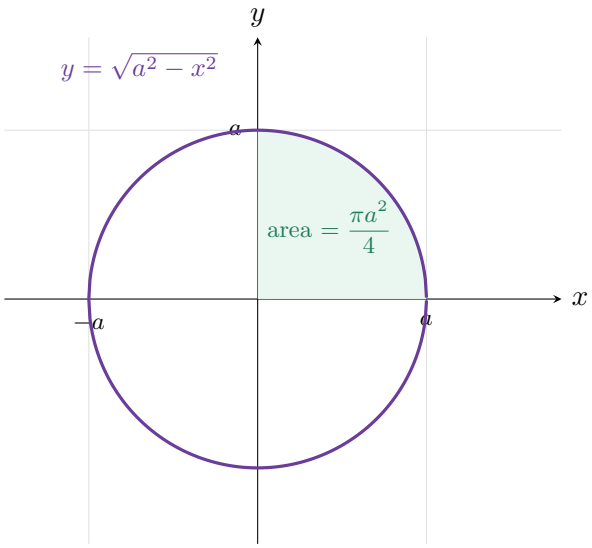
where the $\frac{\sin 2x}{2}$ comes from integrating $\cos 2x$ (integral of cos is sin, and the $2x$ forces a division by 2).

At the lower limit 0 neither term contributes, since $x = 0$ and $\sin 0 = 0$. At the upper limit, $\sin\big(2 \cdot \frac{\pi}{2}\big) = \sin \pi = 0$. Hence

$$\frac{a^2}{2}\left(\frac{\pi}{2} + 0\right) = \boxed{\frac{\pi a^2}{4}}$$

9 The area of a circle — at last, a complete proof

We have seen two ways of computing the area of a circle: **Archimedes'** method of inscribed and circumscribed regular polygons, and the method of cutting the disk into thin circular strips (**annuli**), unrolling them, and observing that in the limit the figure becomes a triangle. Both were very nice — but both involved computing limits, over which we waved our hands a little. Now we use **integrals**.



Draw the axes through the centre — note that no such lines were given to us; all we need is the **radius** a. Consider the part of the circle in the **first quadrant** and think of that arc as a **function** defined on the interval $[0, a]$.

For that we use the equation of the circle: a point lies on it exactly when

$$x^2 + y^2 = a^2,$$

and in the first quadrant both x and y are positive, so we may rewrite this as

$$y = \sqrt{a^2 - x^2}.$$

So consider $f(x) = \sqrt{a^2 - x^2}$ and ask for

$$\int_0^a f(x)\,dx,$$

which is exactly the shaded quarter — the area under the curve, the curve being the graph of this function, namely the arc.

But this is precisely the integral we computed in Example 6.5:

$$\int_0^a \sqrt{a^2 - x^2}\,dx = \frac{\pi a^2}{4}.$$

The area of the whole circle is 4 times this quarter:

$$4 \times \frac{\pi a^2}{4} = \boxed{\pi a^2}$$

and writing the radius as r instead of a gives the familiar πr^2 — of which we now have a **complete proof**.

Integrals are a powerful tool for many things — you will meet them in statistics in particular, but they are also useful in more basic geometric operations, especially in computing areas. You are strongly urged to try computing, for instance, the area of an **ellipse**, or the area under a **hyperbola**.

References and Acknowledgement

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